

Module-2

Session-1

Q1. Write a short paragraph in your own words defining what a 'program' is and give one real-life example from any app you use daily (like Instagram, Zomato, or Paytm).

Ans. A program is a set of instructions given to a computer or mobile phone to perform a specific task.

Programs help us use different apps and complete our daily work easily. For example, when we use **Zomato**.

a program allows us to search for restaurants, choose food, add items to the cart, place the order, make payment, and track the delivery.

Q2. List three popular programming languages you have heard of, and for each, mention whether it is usually compiled, interpreted, or can be both.

***Hint:** Think of languages like Java, Python, and JavaScript.*

Ans. **Java – Usually Compiled**

Java is a popular programming language used to create mobile apps, software, and web applications. Java programs are usually **compiled** into bytecode before they are executed.

Python – Usually Interpreted

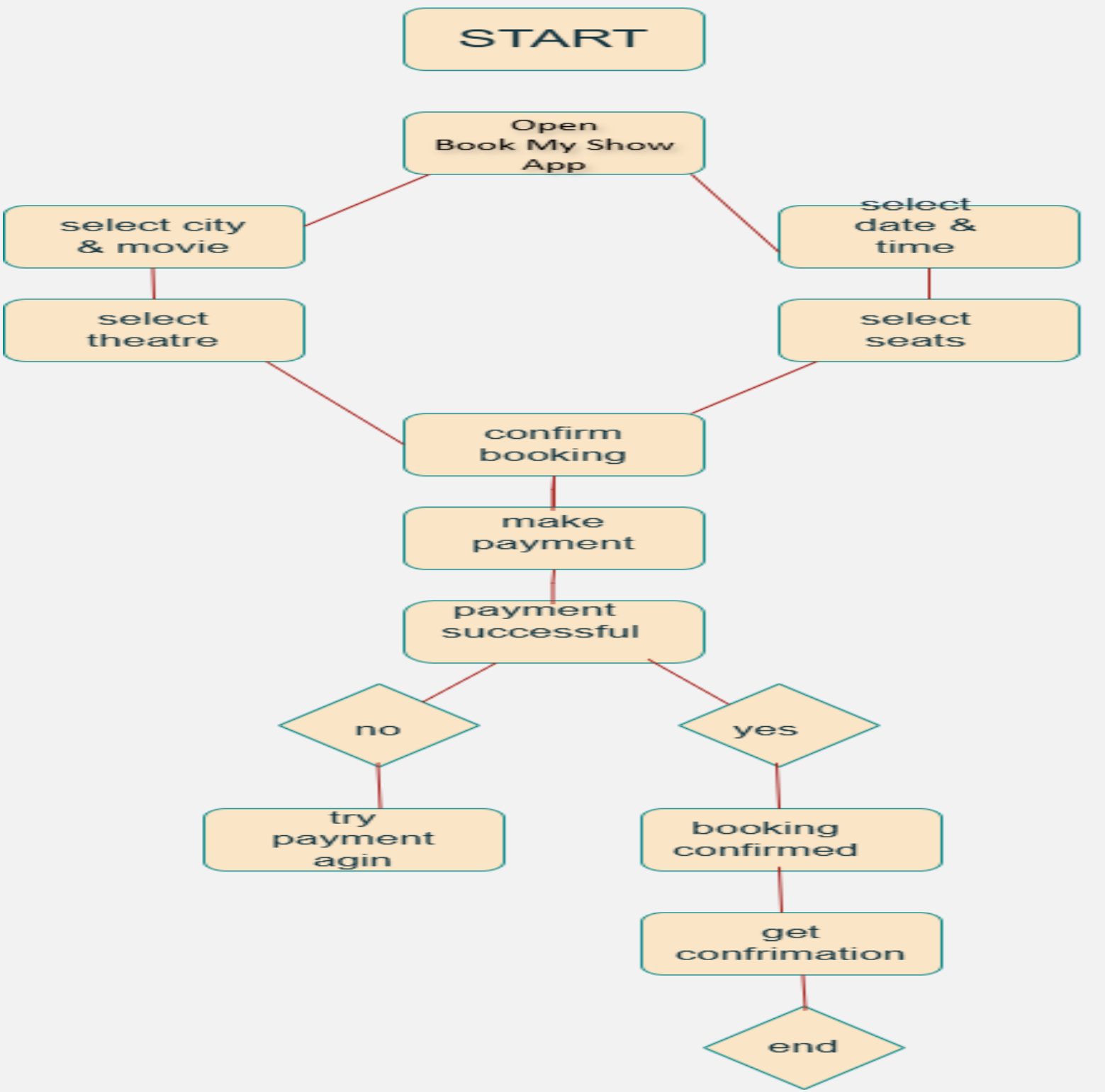
Python is a simple and popular programming language. It is used in websites, data science, artificial intelligence, and automation. Python is usually considered an **interpreted** language.

JavaScript – Can Be Both

JavaScript is mainly used to make websites interactive. It is also used in web applications and some mobile and desktop applications. Modern JavaScript engines can use both interpretation and compilation techniques.

Q3. Draw a flowchart (on paper or using any online tool) for the process of booking a movie ticket on Book My Show, starting from opening the app to receiving the booking confirmation.

Ans.



Q4. Write down the steps (as an algorithm, using simple numbered instructions) for ordering food on Swiggy — from opening the app to completing payment.

Ans. **Start**

Open the **Swiggy app**.

Select your **location**.

Search for a **restaurant or food item**.

Select the food you want.

Add the food to the **cart**.

Open the cart and **check your order**.

Enter or confirm the **delivery address**.

Select a **payment method**.

Make the **payment**.

Payment successful → **Order Confirmed**.

End.

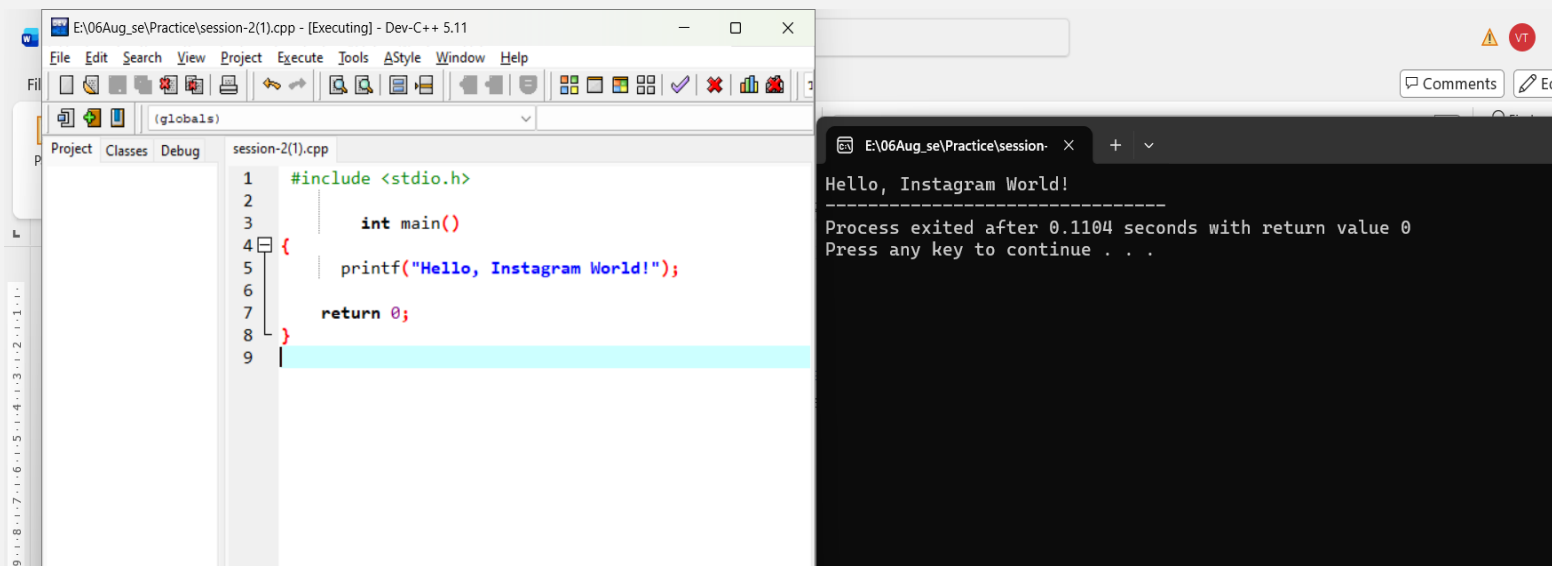
Session-2

Q1. Install GCC on your computer and verify the installation by running `gcc --version` in your terminal or command prompt.

Ans. Install GCC on your computer. Open **Command Prompt/Terminal** and type:

```
gcc --version
```

Q2. Open VS Code or Dev C++ and create a new file named `hello.c`. Write a C program that prints 'Hello, Instagram World!' to the console.



The screenshot displays the Dev-C++ IDE interface. The main editor window shows a C program named `session-2(1).cpp` with the following code:

```
1 #include <stdio.h>
2
3     int main()
4 {
5     printf("Hello, Instagram World!");
6
7     return 0;
8 }
9
```

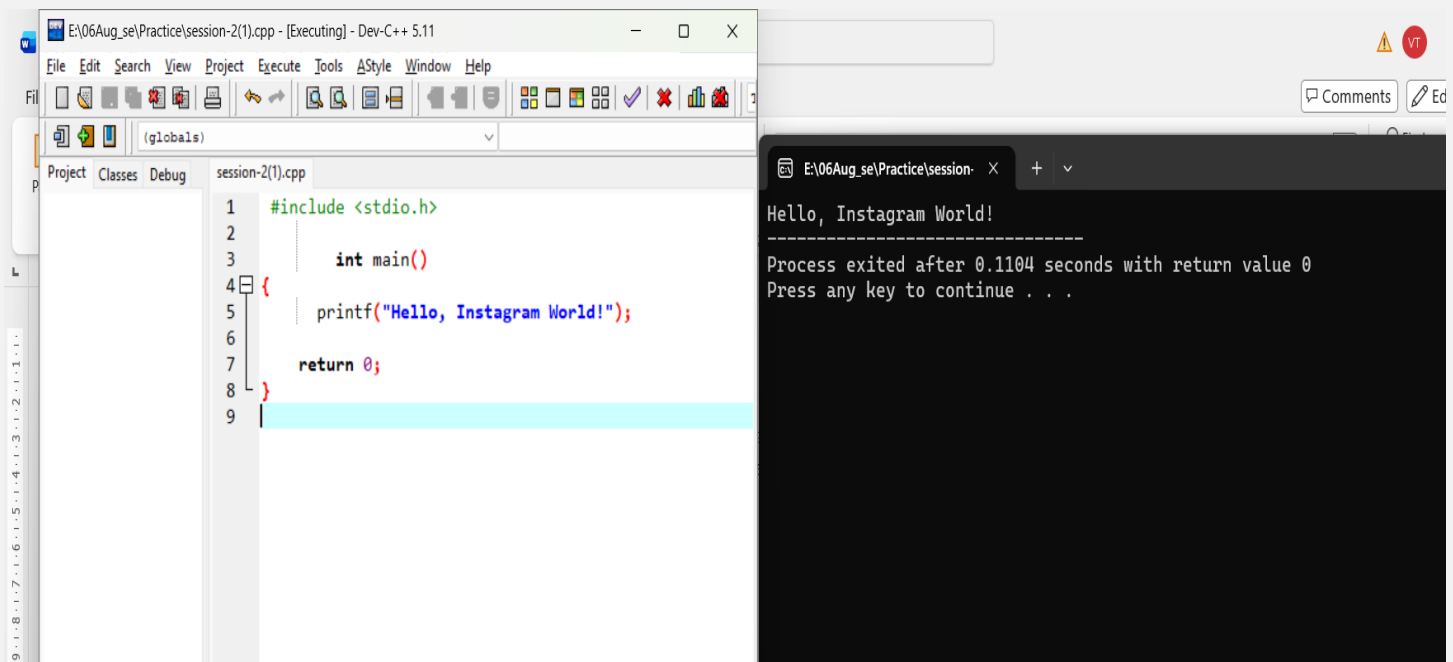
The program is executed, and the output is shown in the console window on the right. The output reads:

```
Hello, Instagram World!
-----
Process exited after 0.1104 seconds with return value 0
Press any key to continue . . .
```

Hello, Instagram World!

Q3. Identify and label the header, main function, and return statement in your hello.c program using comments.

Ans



The screenshot displays the Dev-C++ IDE interface. The left pane shows the source code for 'session-2(1).cpp' with the following content:

```
1  #include <stdio.h>
2
3      int main()
4  {
5      printf("Hello, Instagram World!");
6
7      return 0;
8  }
9
```

The right pane shows the execution output, which includes the program's output and system messages:

```
Hello, Instagram World!
-----
Process exited after 0.1104 seconds with return value 0
Press any key to continue . . .
```

Header: #include <stdio.h>

Main Function: int main()

Return Statement: return 0;

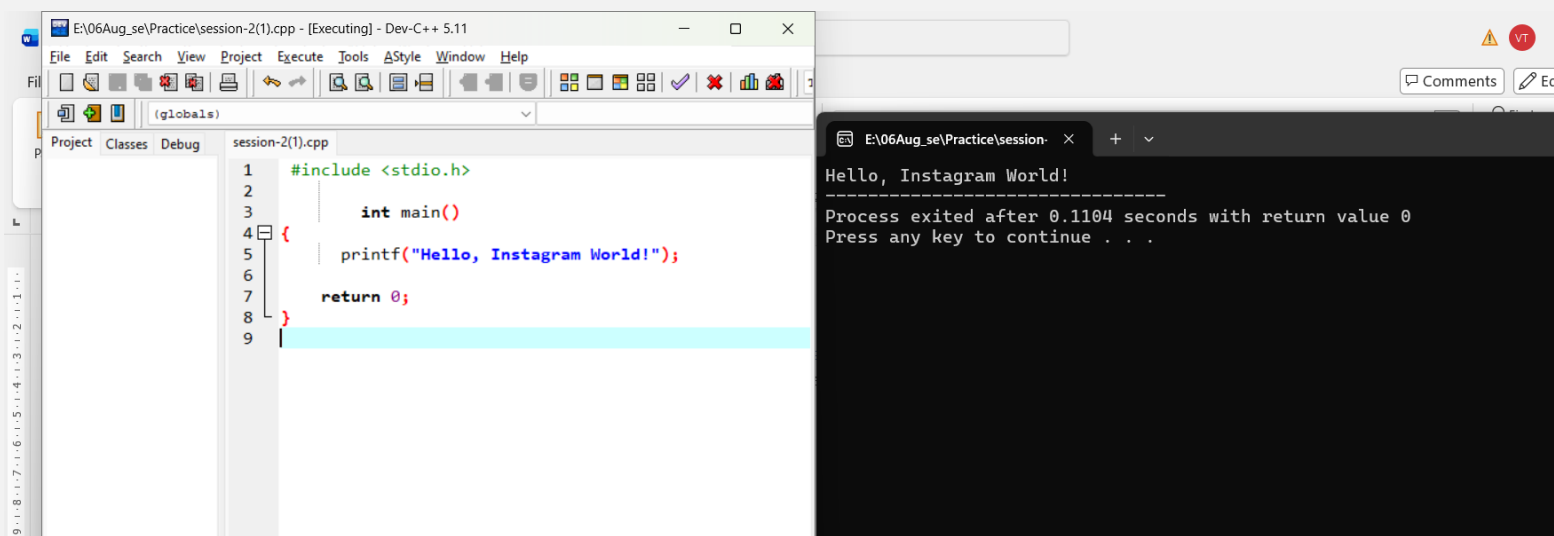
Q4. Compile your hello.c file using the GCC command line (gcc hello.c -o hello) and run the output file to display the message in your terminal.

Hint: On Windows, the output file will be hello.exe; on Linux/Mac, it will be hello.

Ans.

Answer:

- **Header:** #include <stdio.h>
- **Main Function:** int main()
- **Return Statement:** return 0;



The screenshot displays the Dev-C++ IDE interface. The main editor window shows a C program named 'session-2(1).cpp' with the following code:

```
1 #include <stdio.h>
2
3     int main()
4 {
5     printf("Hello, Instagram World!");
6
7     return 0;
8 }
9
```

The code is syntactically highlighted. The line numbers 1 through 9 are visible on the left margin. The 'Project' pane on the left shows a tree view with 'session-2(1).cpp' selected. The 'Output' window on the right shows the execution results:

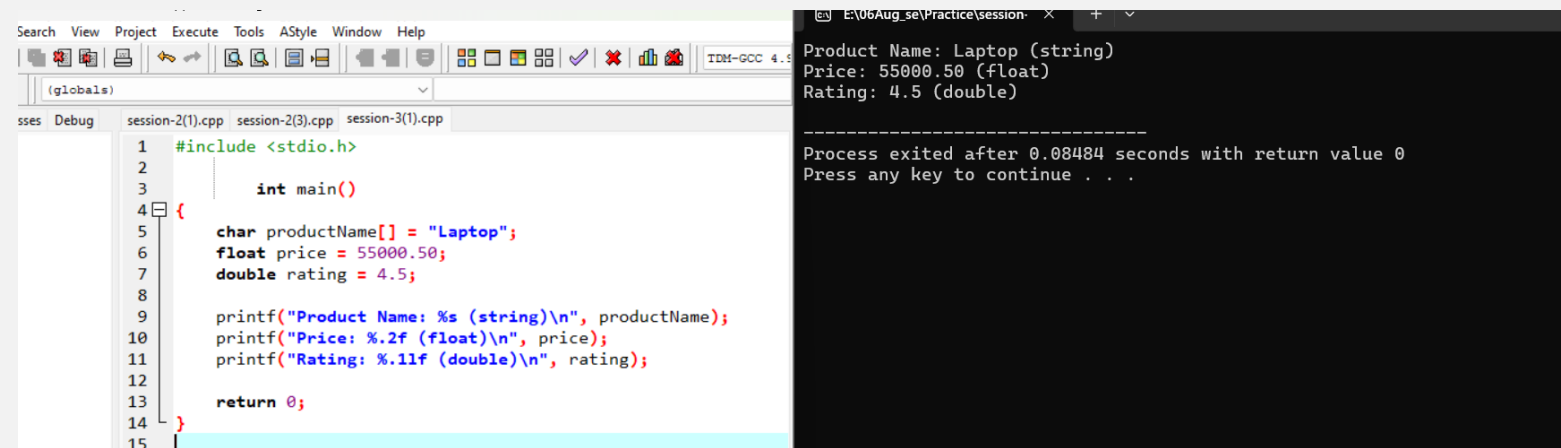
```
Hello, Instagram World!
-----
Process exited after 0.1104 seconds with return value 0
Press any key to continue . . .
```

The output window also shows a tab for 'E:\06Aug_se\Practice\session-2(1).cpp' and a 'Comments' button.

Session-3

Q1. Declare variables for a Flipkart product: product Name (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.

Ans.



The screenshot shows a C++ IDE with two windows. The left window displays the source code for a program that declares and prints variables for a Flipkart product. The right window shows the output of the program, which prints the product name, price, and rating with their respective data types.

```
#include <stdio.h>

int main()
{
    char productName[] = "Laptop";
    float price = 55000.50;
    double rating = 4.5;

    printf("Product Name: %s (string)\n", productName);
    printf("Price: %.2f (float)\n", price);
    printf("Rating: %.11f (double)\n", rating);

    return 0;
}
```

```
Product Name: Laptop (string)
Price: 55000.50 (float)
Rating: 4.5 (double)

-----
Process exited after 0.08484 seconds with return value 0
Press any key to continue . . .
```

Product Name: Laptop (string)

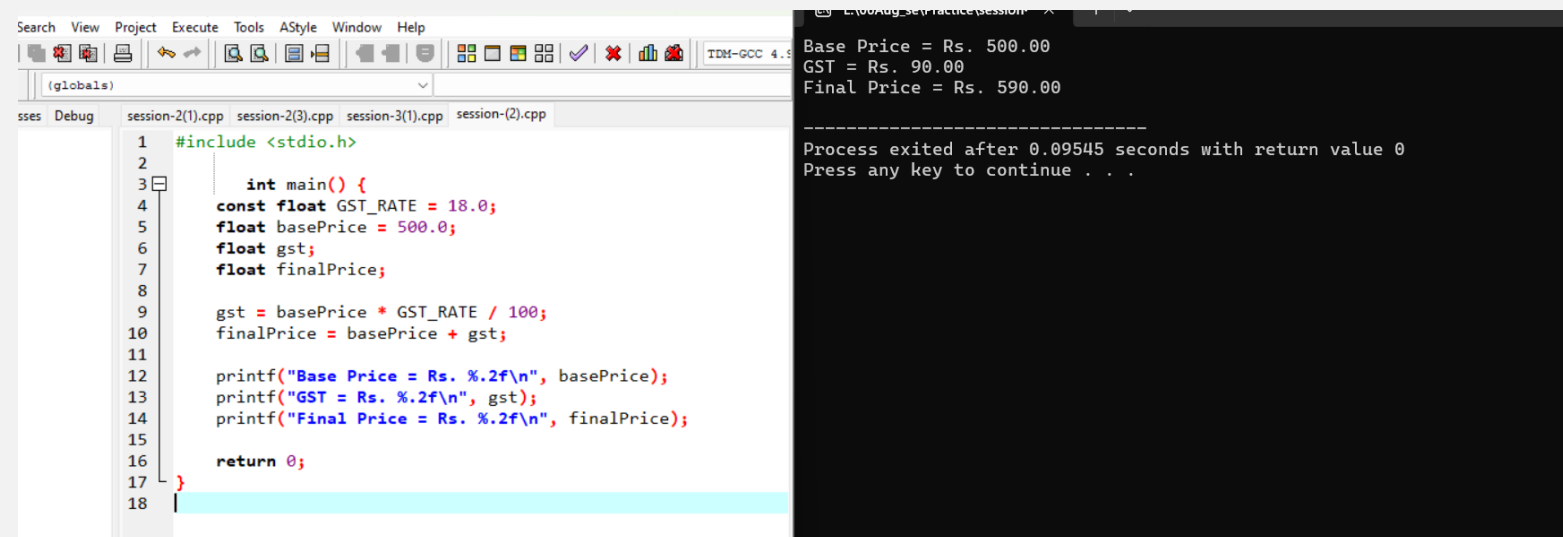
Price: 55000.50 (float)

Rating: 4.5 (double)

Q2. Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.

Constraint: The GST rate must not be changeable after its initial assignment.

Ans



The screenshot shows a C++ IDE with a code editor on the left and a terminal on the right. The code in the editor defines a constant GST rate of 18% and calculates the final price of a Zomato order with a base price of 500.00. The terminal output shows the calculated values: Base Price = Rs. 500.00, GST = Rs. 90.00, and Final Price = Rs. 590.00.

```
1 #include <stdio.h>
2
3 int main() {
4     const float GST_RATE = 18.0;
5     float basePrice = 500.0;
6     float gst;
7     float finalPrice;
8
9     gst = basePrice * GST_RATE / 100;
10    finalPrice = basePrice + gst;
11
12    printf("Base Price = Rs. %.2f\n", basePrice);
13    printf("GST = Rs. %.2f\n", gst);
14    printf("Final Price = Rs. %.2f\n", finalPrice);
15
16    return 0;
17 }
18
```

```
Base Price = Rs. 500.00
GST = Rs. 90.00
Final Price = Rs. 590.00

-----
Process exited after 0.09545 seconds with return value 0
Press any key to continue . . .
```

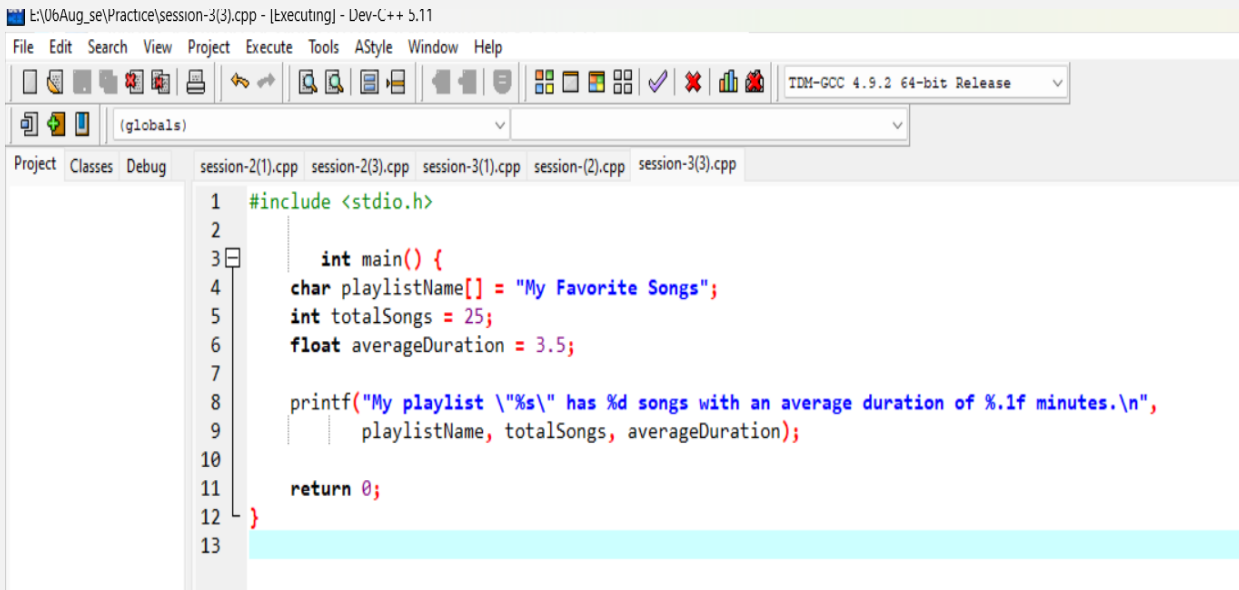
Base Price = ₹500

GST = ₹90

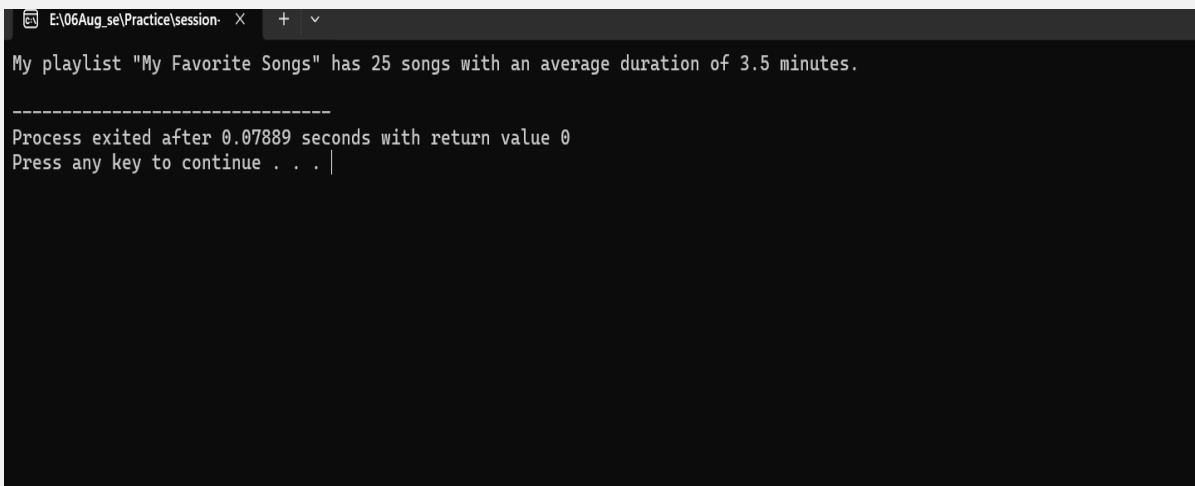
Final Price = ₹590

Q3. Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.

Ans.



```
1 #include <stdio.h>
2
3 int main() {
4     char playlistName[] = "My Favorite Songs";
5     int totalSongs = 25;
6     float averageDuration = 3.5;
7
8     printf("My playlist \"%s\" has %d songs with an average duration of %.1f minutes.\n",
9           playlistName, totalSongs, averageDuration);
10
11     return 0;
12 }
13
```



```
My playlist "My Favorite Songs" has 25 songs with an average duration of 3.5 minutes.
-----
Process exited after 0.07889 seconds with return value 0
Press any key to continue . . .
```

Playlist Name → My Favorite Songs

- Total Songs → 25
- Average Duration → 3.5 minutes

Q4. Identify and correct the invalid variable names from this list: 1stPlayer, player_age, \$score, total-marks, userName.

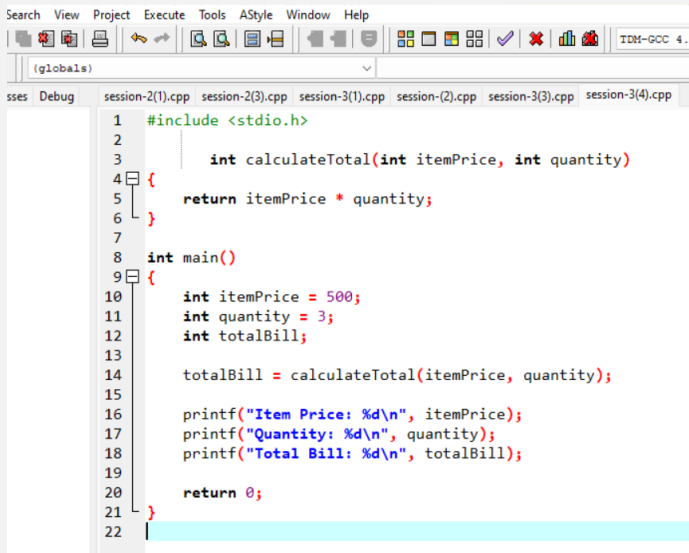
Hint: Apply standard variable naming rules for your programming language.

Ans. 1stPlayer	→	firstPlayer
player_age	→	player_age
\$score	→	score
total-marks	→	total_marks
userName	→	userName

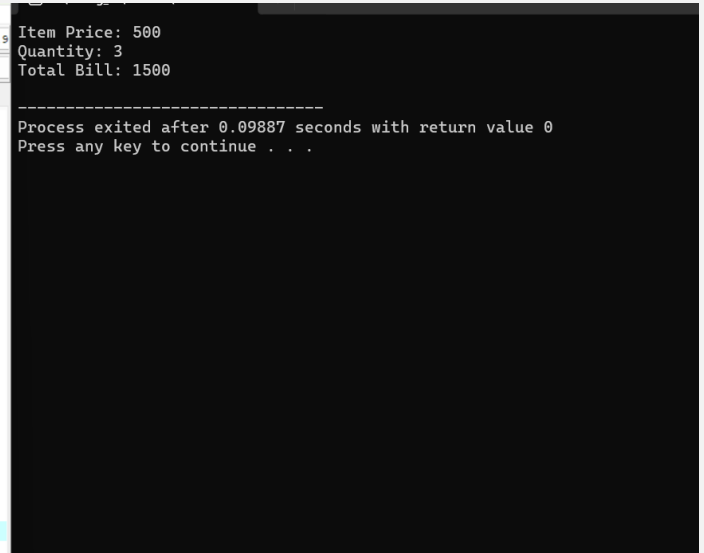
Session-4

Q1. Create a simple JavaScript function called `calculateTotal` that takes two numbers: `itemPrice` and `quantity`, and returns the total bill amount using arithmetic operators.

Ans.



```
1 #include <stdio.h>
2
3     int calculateTotal(int itemPrice, int quantity)
4 {
5     return itemPrice * quantity;
6 }
7
8 int main()
9 {
10     int itemPrice = 500;
11     int quantity = 3;
12     int totalBill;
13
14     totalBill = calculateTotal(itemPrice, quantity);
15
16     printf("Item Price: %d\n", itemPrice);
17     printf("Quantity: %d\n", quantity);
18     printf("Total Bill: %d\n", totalBill);
19
20     return 0;
21 }
22
```



```
Item Price: 500
Quantity: 3
Total Bill: 1500

-----
Process exited after 0.09887 seconds with return value 0
Press any key to continue . . .
```

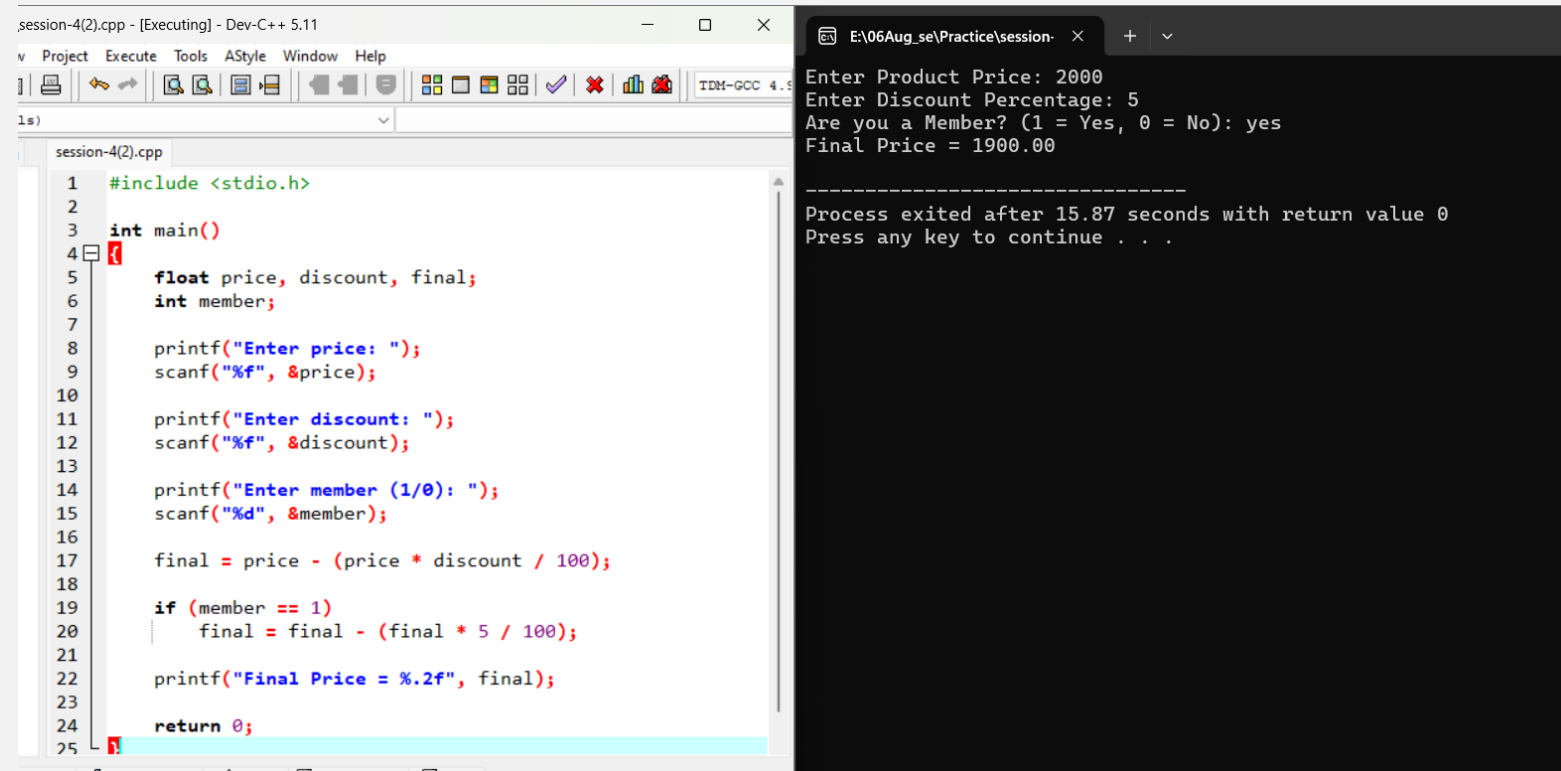
Item Price: 500

Quantity: 3

Total Bill: 1500

Q2. Build a Flipkart-style discount calculator: given product price, discount percentage, and a boolean is Member, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if isMember is true).

Ans.



The image shows a screenshot of a C++ IDE (Dev-C++ 5.11) with a file named 'session-4(2).cpp'. The code is as follows:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float price, discount, final;
6     int member;
7
8     printf("Enter price: ");
9     scanf("%f", &price);
10
11    printf("Enter discount: ");
12    scanf("%f", &discount);
13
14    printf("Enter member (1/0): ");
15    scanf("%d", &member);
16
17    final = price - (price * discount / 100);
18
19    if (member == 1)
20        final = final - (final * 5 / 100);
21
22    printf("Final Price = %.2f", final);
23
24    return 0;
25 }
```

The output window on the right shows the execution results:

```
Enter Product Price: 2000
Enter Discount Percentage: 5
Are you a Member? (1 = Yes, 0 = No): yes
Final Price = 1900.00

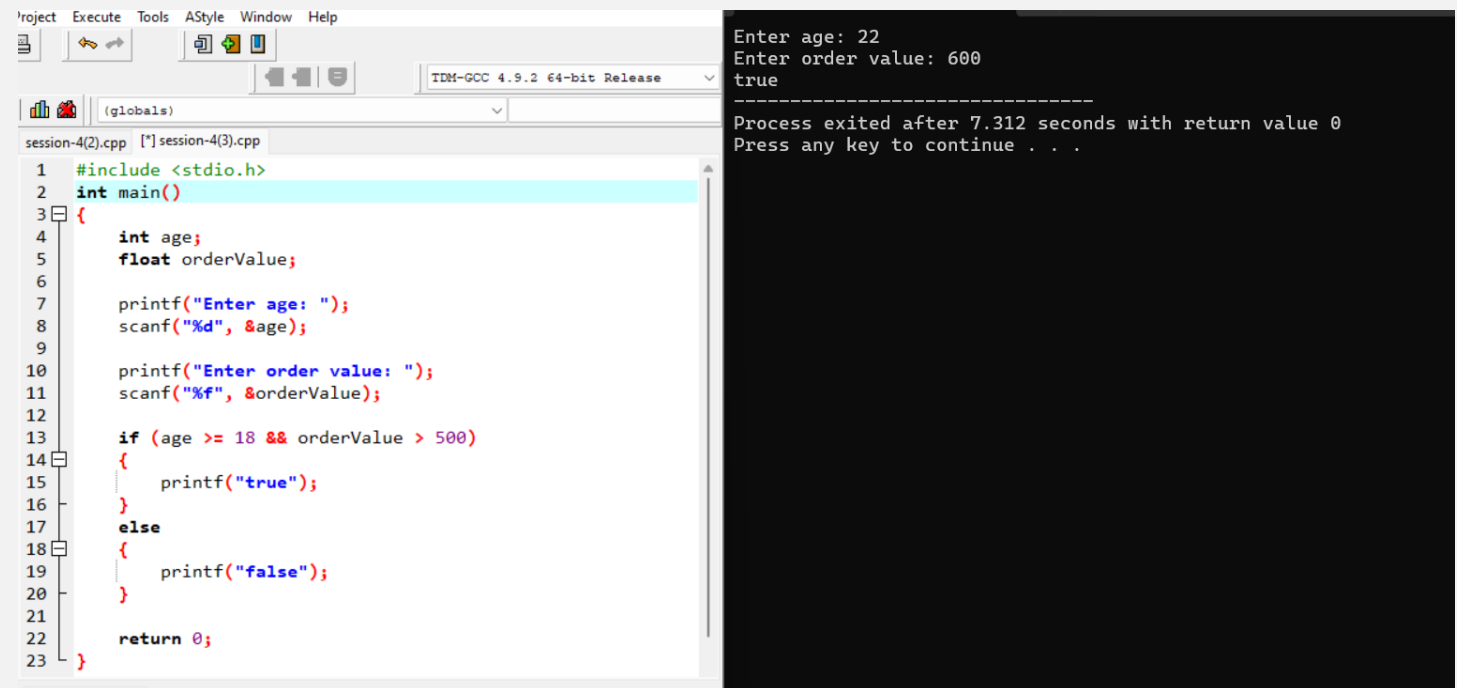
-----
Process exited after 15.87 seconds with return value 0
Press any key to continue . . .
```

Enter Product Price: 2000
Enter Discount Percentage: 5
Are you a Member (1 = Yes, 0 = No): yes
Final Price = 1900.00

Q3. Write a function isEligibleForOffer that takes a user's age and total order value, and returns true if the user is 18 or older AND the order value is above 500, otherwise false.

Hint: Use relational and logical operators together.

Ans.



The screenshot shows a C++ IDE with a source code editor on the left and a terminal window on the right. The source code defines a function `isEligibleForOffer` that takes `age` and `orderValue` as parameters and returns a boolean. The `main` function prompts the user for age and order value, then calls `isEligibleForOffer` and prints the result. The terminal output shows the user entering age 22 and order value 600, resulting in the output `true`.

```
#include <stdio.h>

int main()
{
    int age;
    float orderValue;

    printf("Enter age: ");
    scanf("%d", &age);

    printf("Enter order value: ");
    scanf("%f", &orderValue);

    if (age >= 18 && orderValue > 500)
    {
        printf("true");
    }
    else
    {
        printf("false");
    }

    return 0;
}
```

```
Enter age: 22
Enter order value: 600
true
-----
Process exited after 7.312 seconds with return value 0
Press any key to continue . . .
```

Enter age: 22

Enter order value: 600

True

Q4. Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.

Ans. #include <stdio.h>

```
int main()
```

```
{
```

```
    int likes, comments, shares;
```

```
    printf("Enter likes: ");
```

```
scanf("%d", &likes);
```

```
printf("Enter comments: ");
```

```
scanf("%d", &comments);
```

```
printf("Enter shares: ");
```

```
scanf("%d", &shares);
```

```
if (likes >= 1000 || (comments > 200 && shares >= 50))
```

```
{
```

```
    printf("Trending");
```

```
}
```

```
else
```

```
{
```

```
    printf("Not Trending");
```

```
}
```

```
return 0;
```

```
}
```

Enter likes: 1200

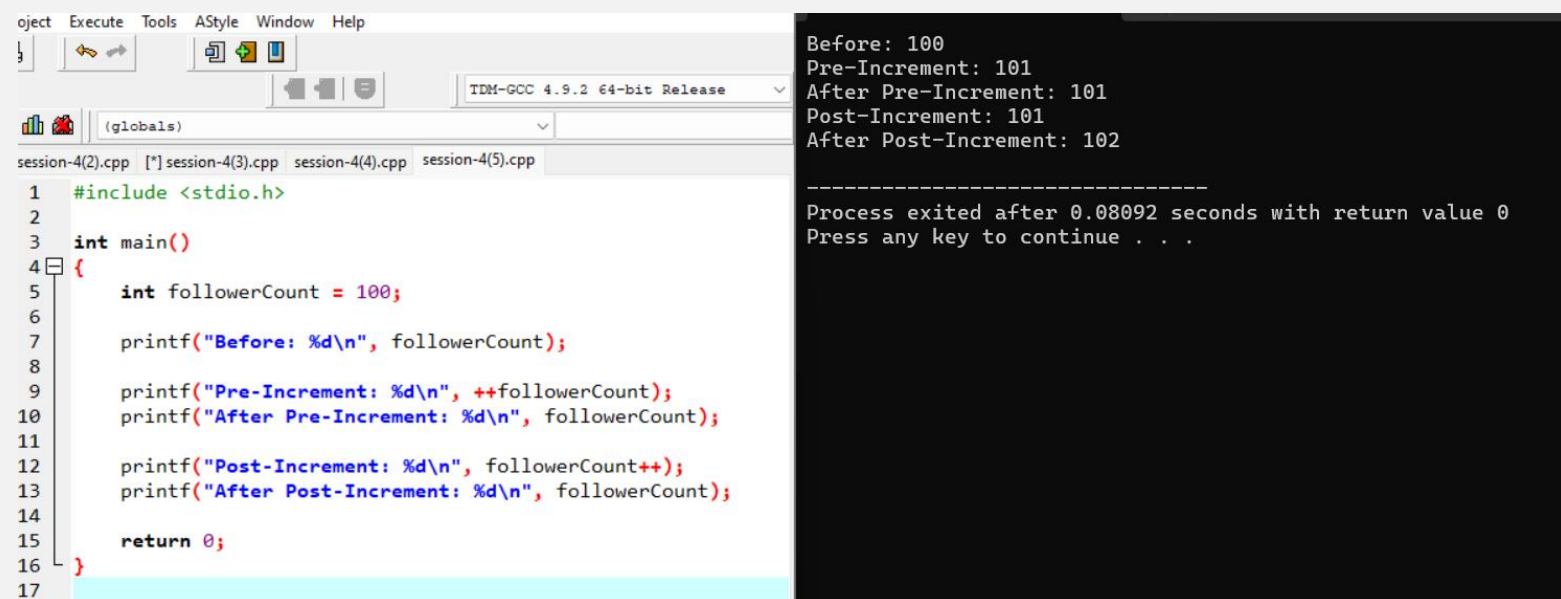
Enter comments: 150

Enter shares: 60

Trending

Q5. Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a follower Count variable.

Ans.

The image shows a screenshot of a C++ IDE. On the left, the code editor displays a C++ program. The code includes `<stdio.h>` and defines a `main` function. Inside `main`, a variable `followerCount` is initialized to 100. It then prints the value before any increments. Next, it performs a pre-increment (`++followerCount`) and prints the value. Then, it performs a post-increment (`followerCount++`) and prints the value. Finally, it returns 0. On the right, the output window shows the execution results: 'Before: 100', 'Pre-Increment: 101', 'After Pre-Increment: 101', 'Post-Increment: 101', and 'After Post-Increment: 102'. Below the output, a message indicates the process exited after 0.08092 seconds with a return value of 0, and prompts the user to press any key to continue.

Before: 100

Pre-Increment: 101

After Pre-Increment: 101

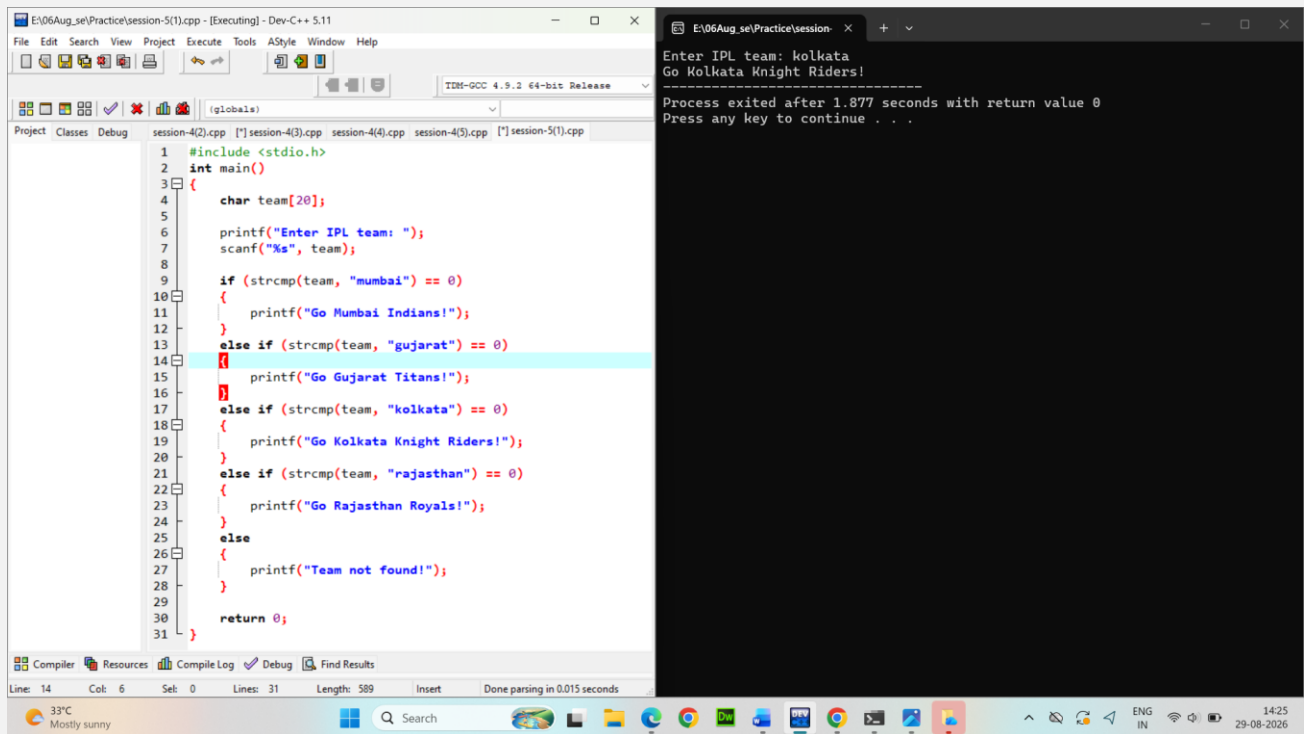
Post-Increment: 101

After Post-Increment: 102

Session-5

Q1. Create a simple IPL Fan Bot that takes your favorite IPL team name as input and uses if-else-if statements to print a unique cheer message for each team (e.g., 'Go Mumbai Indians!', 'Chennai Super Kings for the win!'). If the team is not recognized, print 'Team not found!'

Ans



The screenshot displays a C++ IDE with a source code editor on the left and a terminal window on the right. The source code is as follows:

```
1 #include <stdio.h>
2 int main()
3 {
4     char team[20];
5
6     printf("Enter IPL team: ");
7     scanf("%s", team);
8
9     if (strcmp(team, "mumbai") == 0)
10    {
11        printf("Go Mumbai Indians!");
12    }
13    else if (strcmp(team, "gujarat") == 0)
14    {
15        printf("Go Gujarat Titans!");
16    }
17    else if (strcmp(team, "kolkata") == 0)
18    {
19        printf("Go Kolkata Knight Riders!");
20    }
21    else if (strcmp(team, "rajasthan") == 0)
22    {
23        printf("Go Rajasthan Royals!");
24    }
25    else
26    {
27        printf("Team not found!");
28    }
29
30    return 0;
31 }
```

The terminal window on the right shows the program's execution:

```
Enter IPL team: kolkata
Go Kolkata Knight Riders!
-----
Process exited after 1.877 seconds with return value 0
Press any key to continue . . .
```

Output:

Enter IPL team: kolkata
Go Mumbai Indians!

Q2. Build a Zomato-style food suggestion tool: take the user's preferred meal time ('breakfast', 'lunch', 'dinner', or 'snack') and use a switch-case statement to suggest a popular dish for that time. If the input doesn't match any meal, suggest 'Try some fruits!'.

Ans.

ProjectExecuteToolsAStyleWindowHelp

(globals)

[*] session-5(1).cpp [*] session-5(2).cpp

```
1 #include <stdio.h>
2 int main()
3 {
4     int choice;
5
6     printf("1. Breakfast\n");
7     printf("2. Lunch\n");
8     printf("3. Dinner\n");
9     printf("4. Snack\n");
10
11     printf("Enter your choice: ");
12     scanf("%d", &choice);
13
14     switch(choice)
15     {
16         case 1:
17             printf("Try Dosa!");
18             break;
19
20         case 2:
21             printf("Try Gujarati Thali!");
22             break;
23
24         case 3:
25             printf("Try Pizza!");
26             break;
27
28         case 4:
29             printf("Try Samosa!");
30             break;
31
32         default:
33             printf("Try some fruits!");
34     }
35 }
```

```
1. Breakfast
2. Lunch
3. Dinner
4. Snack
Enter your choice: snack
Try some fruits!
-----
Process exited after 14.59 seconds with return value 0
Press any key to continue . . .
```

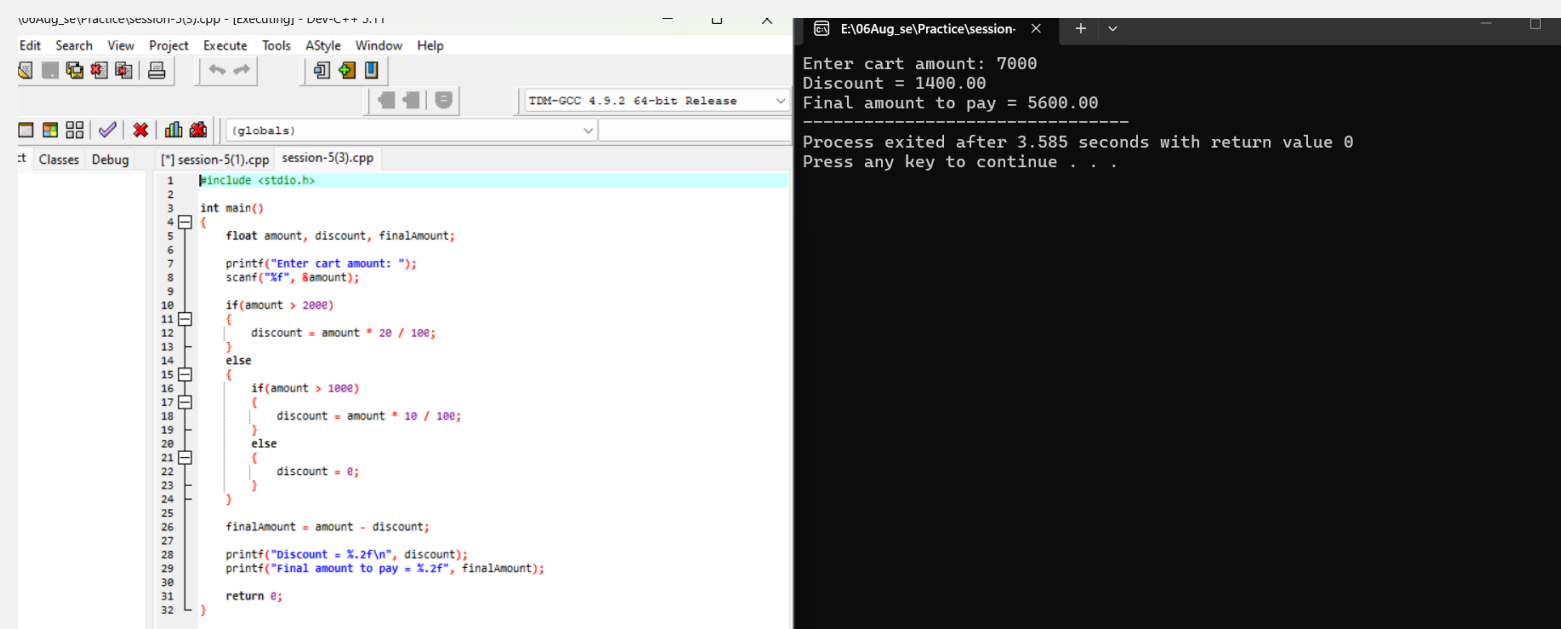
Output

1. Breakfast
2. Lunch
3. Dinner
4. Snack
Enter your choice: 2
Try Gujarati Thali!

Q3. Create a Flipkart discount calculator that asks the user for the total cart amount. Use nested if statements to check: if amount > 2000, apply 20% discount; else if amount > 1000, apply 10% discount; else, no discount. Print the final amount to pay.

Hint: Use nested ifs to check each discount slab.

Ans.



The image shows a C++ IDE with two windows. The left window displays the source code for a discount calculator, and the right window shows the program's output.

```
1 #include <stdio.h>
2
3 int main()
4 {
5     float amount, discount, finalAmount;
6
7     printf("Enter cart amount: ");
8     scanf("%f", &amount);
9
10    if(amount > 2000)
11    {
12        discount = amount * 20 / 100;
13    }
14    else
15    {
16        if(amount > 1000)
17        {
18            discount = amount * 10 / 100;
19        }
20        else
21        {
22            discount = 0;
23        }
24    }
25
26    finalAmount = amount - discount;
27
28    printf("Discount = %.2f\n", discount);
29    printf("Final amount to pay = %.2f", finalAmount);
30
31    return 0;
32 }
```

The output window shows the following text:

```
Enter cart amount: 7000
Discount = 1400.00
Final amount to pay = 5600.00
-----
Process exited after 3.585 seconds with return value 0
Press any key to continue . . .
```

Output

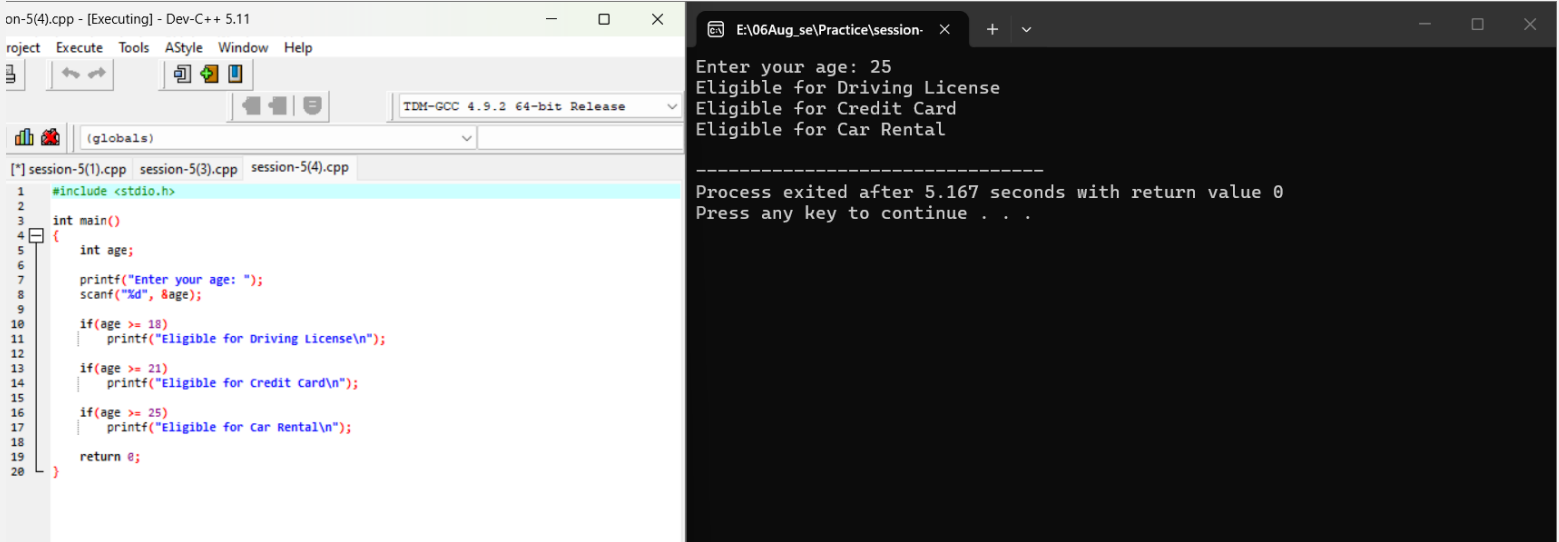
Enter cart amount: 7000

Discount = 1400.00

Final amount to pay = 5600.00

Q4. Write a program that takes a user's age and checks eligibility for three things using if-else statements: if age ≥ 18 , print 'Eligible for Driving License'; if age ≥ 21 , also print 'Eligible for Credit Card'; if age ≥ 25 , also print 'Eligible for Car Rental'. Print all applicable messages for the given age.

Ans



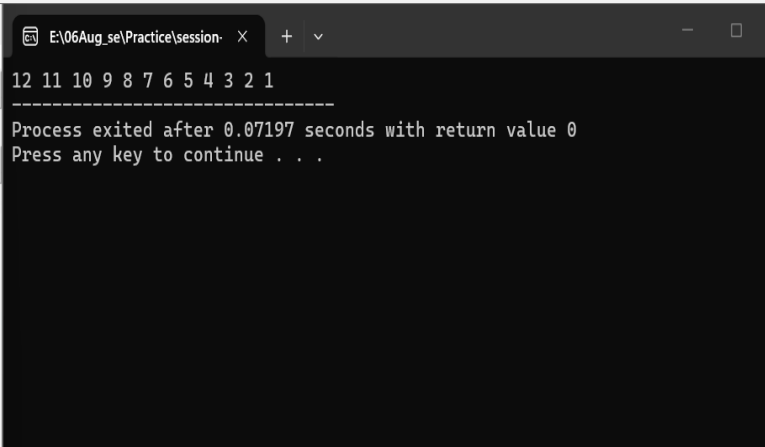
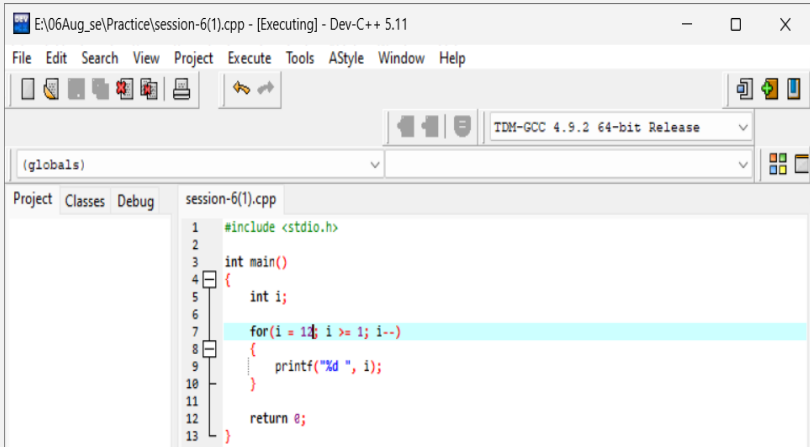
Output

Enter your age: 22
Eligible for Driving License
Eligible for Credit Card

Session-6

Q1. Write a program that prints the numbers from 10 down to 1 using a for loop, similar to a countdown timer.

Ans.

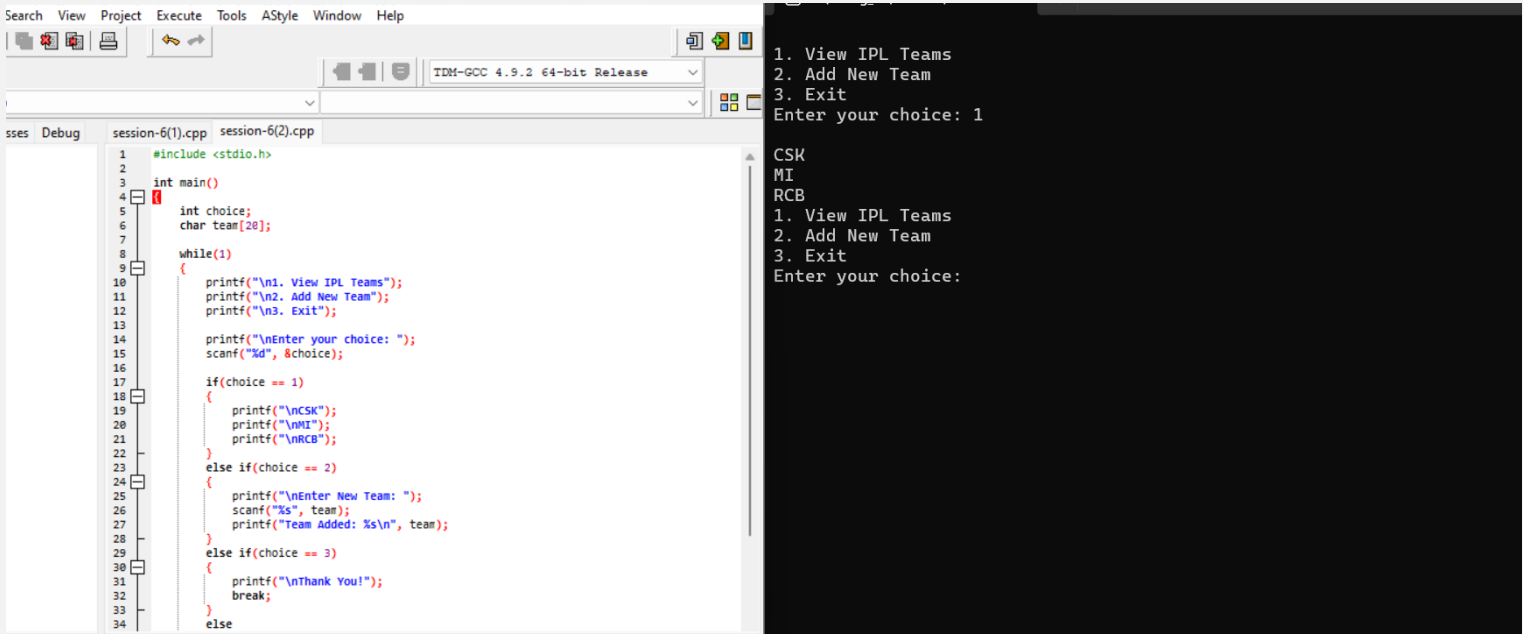


12 11 10 9 8 7 6 5 4 3 2 1

Q2. Create a menu-driven console app that lets the user: 1) View your favorite 3 IPL teams, 2) Add a new team, 3) Exit. Use a while loop to keep showing the menu until the user chooses Exit .

Hint: Use input() (or Scanner in Java) to get the user's choice each time.

Ans



View ipl teams
Add new team
Exit
Enter your choice:1
Csk
Mi
rcb

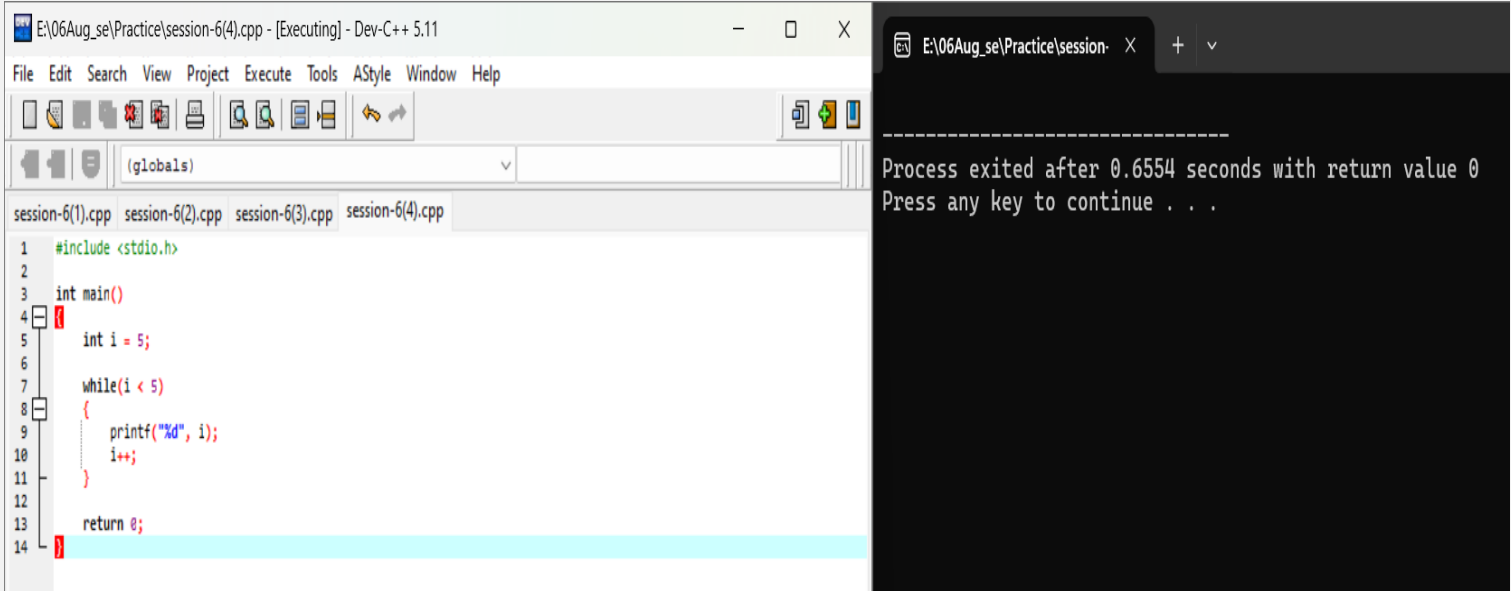
Q3. Build a 'Guess the Song' game like Spotify — the program randomly picks a song name from a list and asks the user to guess it. Use a do-while loop so the user can keep guessing until they get it right.

Constraint: Use at least 3 song names of your choice.

Ans.

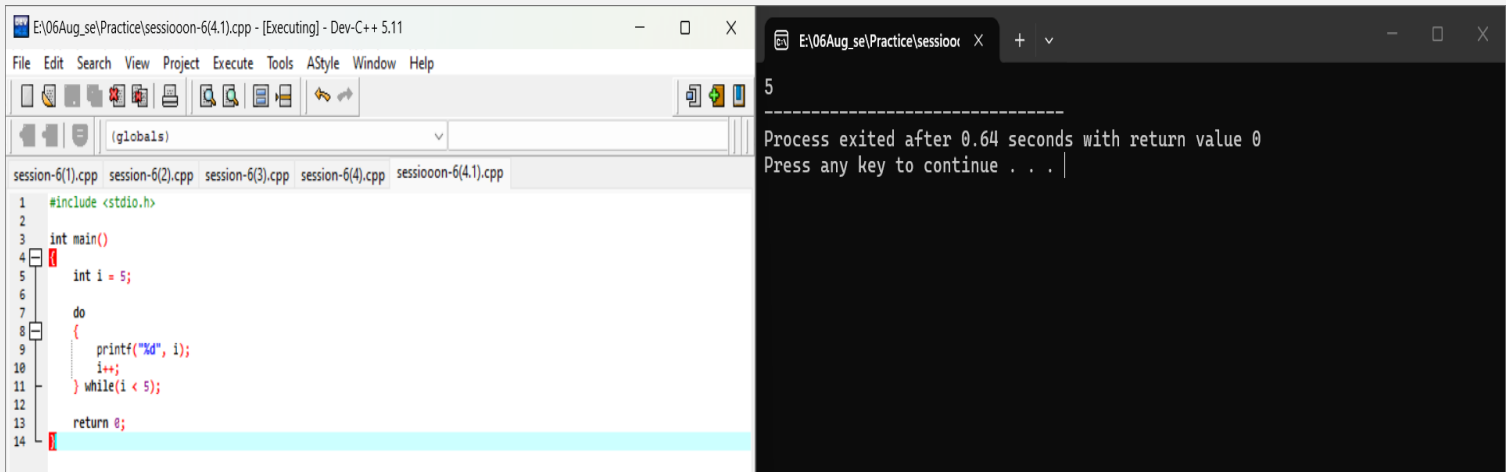
Q4. Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

Ans. Entry-controlled



No Output

Exit -controlled



Session-7

Q1. Use nested for loops to print a grid of emojis representing a 5x5 Instagram post feed, where each cell shows a 📷 symbol.

Ans.

E:\06Aug_se\Practice\session-7(1).cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

(globals)

session-6(1).cpp session-6(2).cpp session-6(3).cpp session-6(4).cpp session-6(4.1).cpp session-7(1).cpp

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int i, j;
6
7     for(i = 1; i <= 5; i++)
8     {
9         for(j = 1; j <= 5; j++)
10        {
11            printf(" #");
12        }
13        printf("\n");
14    }
15
16    return 0;
17 }
```

E:\06Aug_se\Practice\session-7(1).cpp

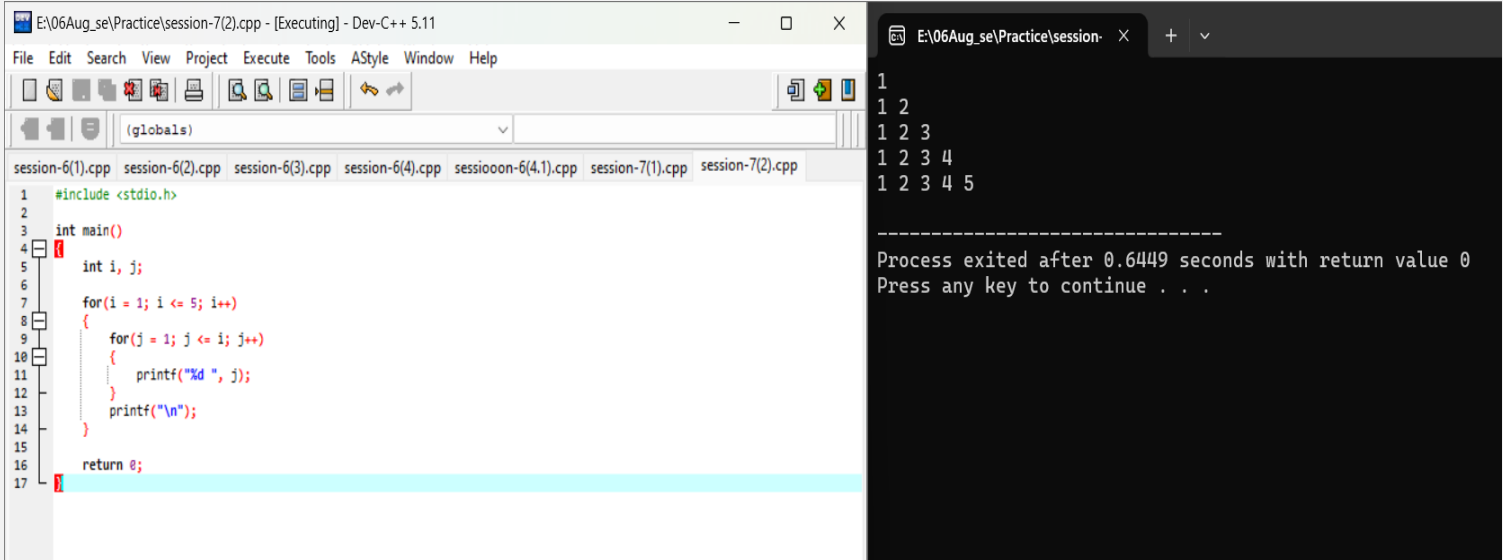
```
# # # # #
# # # # #
# # # # #
# # # # #
# # # # #

-----
Process exited after 0.6816 seconds with return value 0
Press any key to continue . . .
```


#

Q2. Build a right-angled triangle pattern using nested loops, where each row displays increasing numbers starting from 1, similar to how a leaderboard on a gaming app shows rank numbers.

Ans.

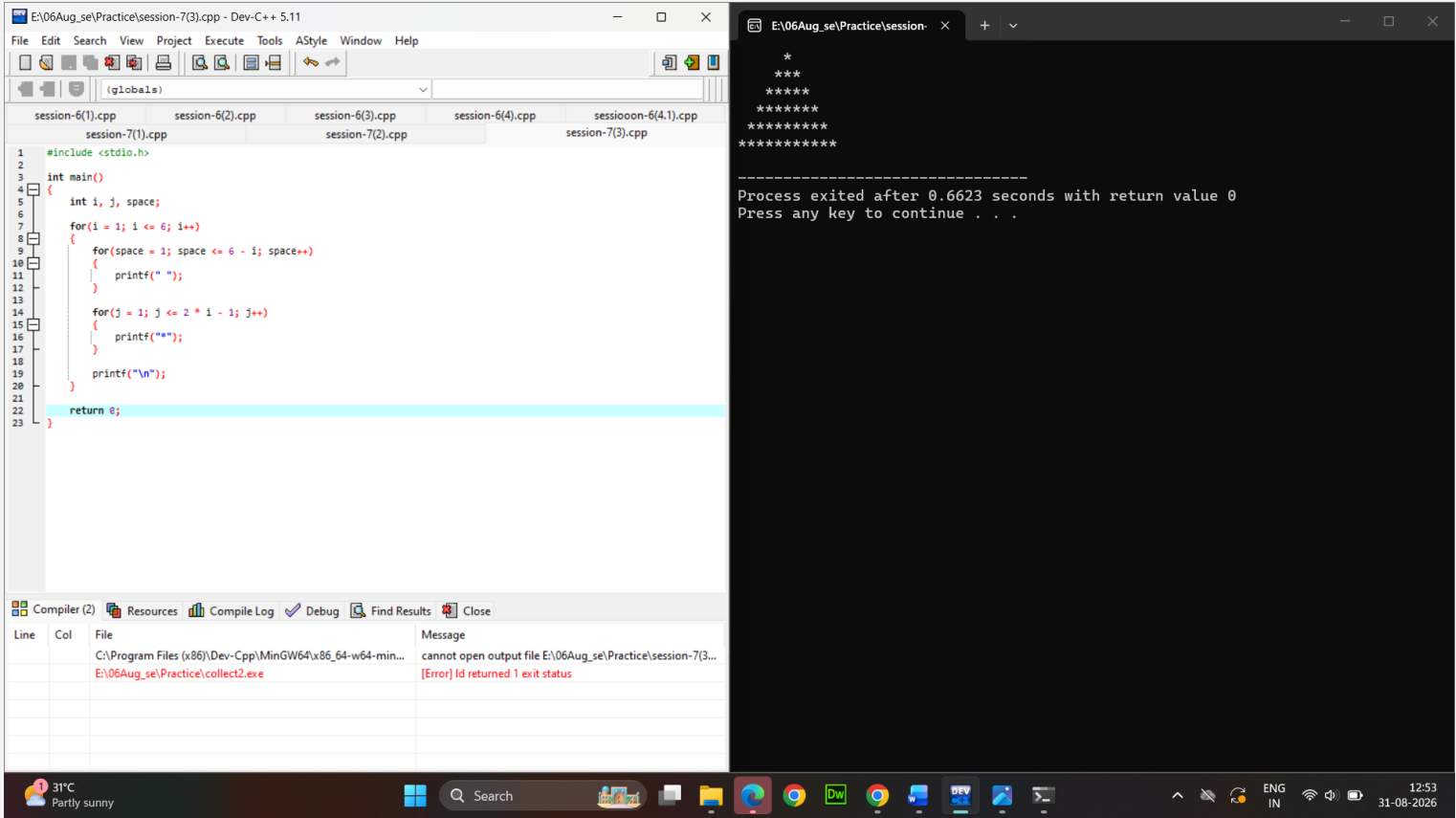


1
12
123
1234
12345

Q3. Create a pattern that prints a pyramid of stars (*) with 6 rows, centered like the loading animation you see on Book My Show when a page is loading.

Hint: Use spaces to align the stars in the center for each row.

Ans.



```

*

* * *

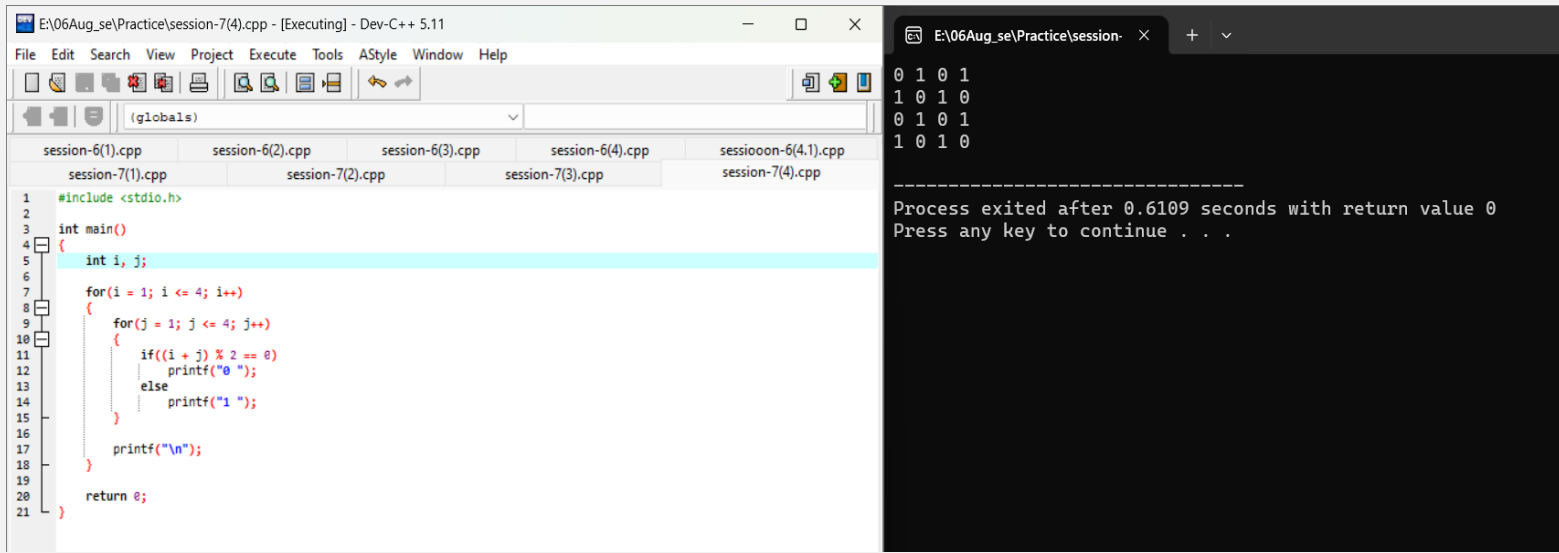
* * * * *

* * * * *

```

Q4. Write code using nested loops to print a pattern of alternating 0s and 1s in a grid, like the checkered background seen in some Spotify playlist covers (e.g., for a 4x4 grid, alternate 0 and 1 in each cell)

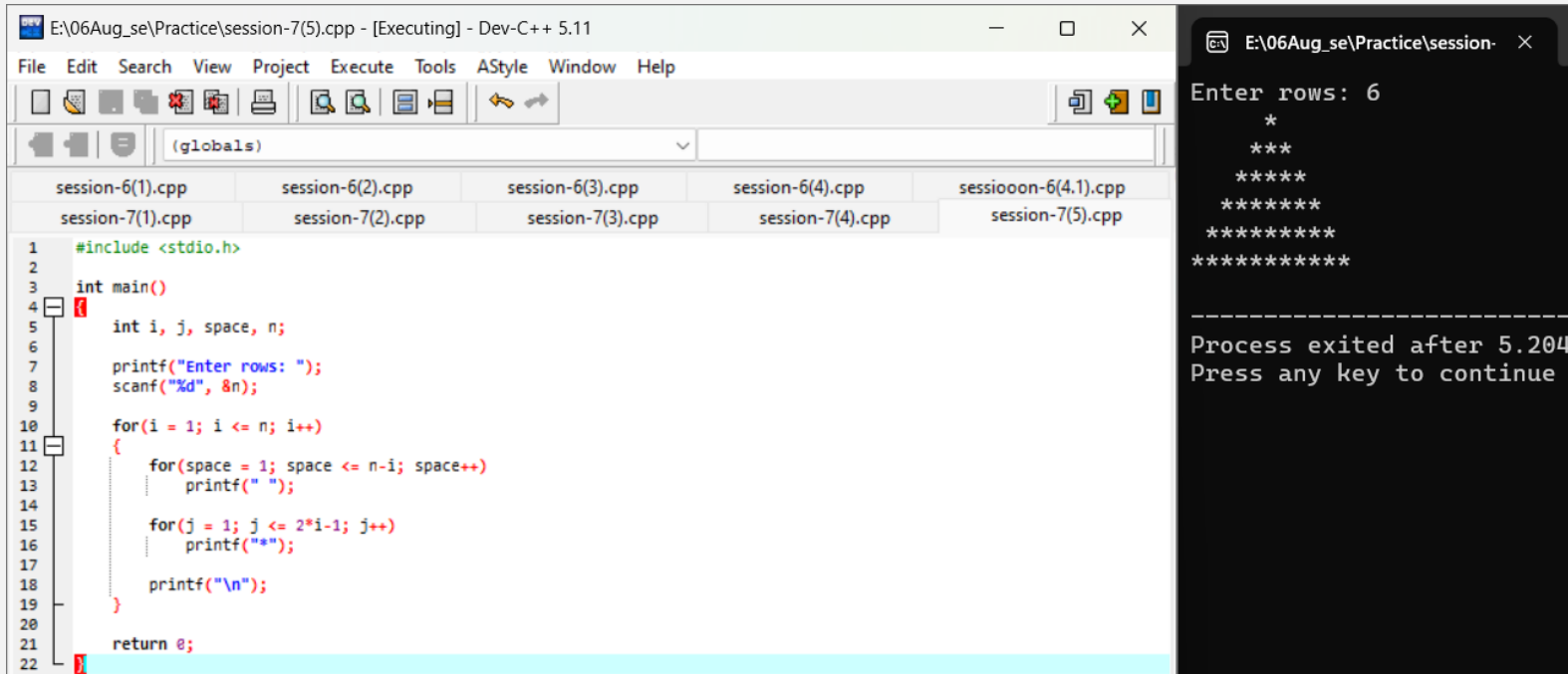
Ans.



0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 0

Q 5. Modify your pyramid pattern code to accept the number of rows as user input, so the user can set the height of the pyramid before printing.

Ans.



Enter row: 6

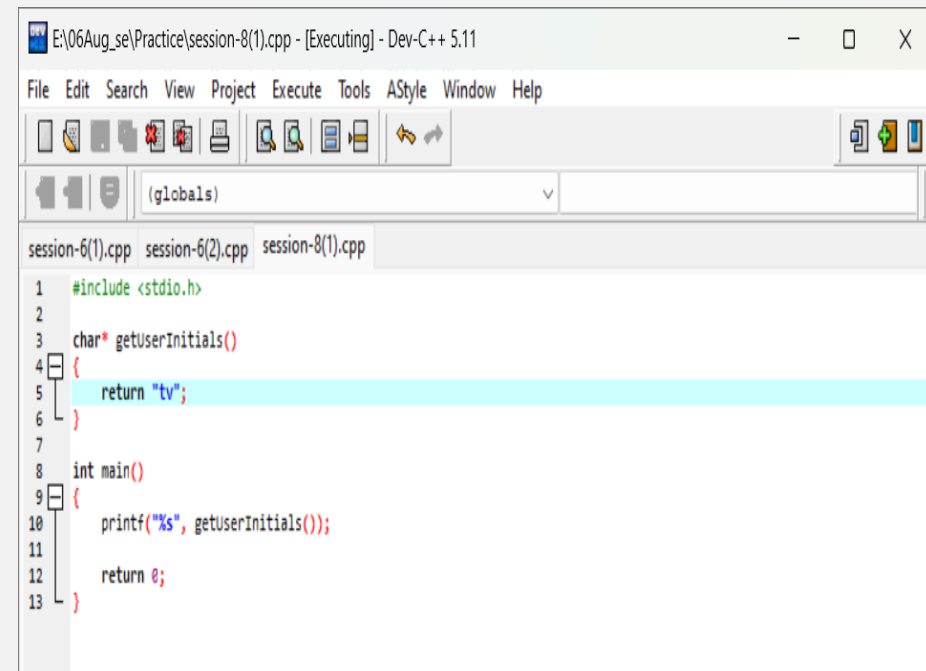
```

      *
    * * *
  * * * * *
* * * * * *
* * * * * * *
* * * * * * *
```

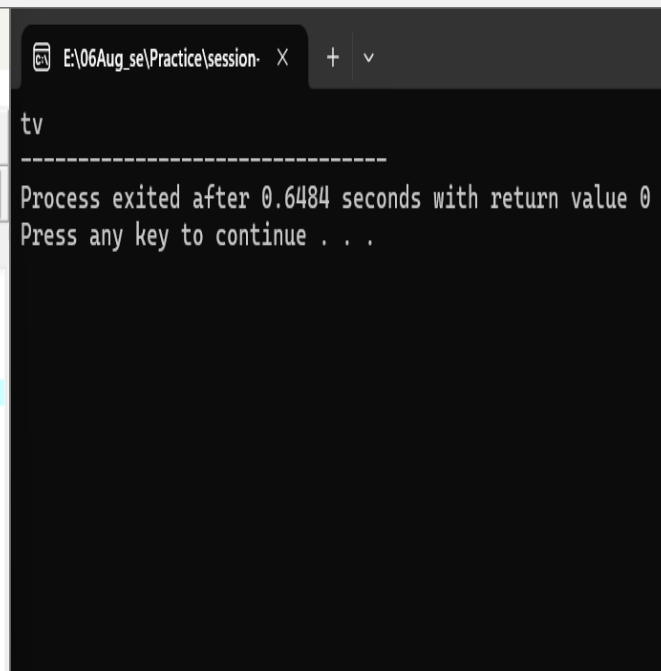
Session-8

Q 1. Declare a function called `getUserInitials` that takes a user's full name (like 'Virat Kohli') and returns their initials in uppercase (e.g., 'VK'). Call this function with your favorite cricketer's name and print the result.

Ans.



```
1  #include <stdio.h>
2
3  char* getUserInitials()
4  {
5      return "tv";
6  }
7
8  int main()
9  {
10     printf("%s", getUserInitials());
11
12     return 0;
13 }
```



```
tv
-----
Process exited after 0.6484 seconds with return value 0
Press any key to continue . . .
```

Q 2. Create a function add To Cart that takes a shopping cart array and a product name, adds the product to the cart, and prints the updated cart. Demonstrate how passing the cart array by reference allows changes to persist outside the function.

Hint: In languages like JavaScript, arrays are passed by reference. In C/C++, use pointers for reference behavior.

Ans.

E:\06Aug_se\Practice\session-8(2).cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

(globals)

session-6(1).cpp session-6(2).cpp session-8(1).cpp session-8(2).cpp

```
1 #include <stdio.h>
2
3 void addToCart(char *cart[], int *count, char *product)
4 {
5     cart[*count] = product;
6     (*count)++;
7 }
8
9 int main()
10 {
11     char *cart[5];
12     int count = 0;
13
14     addToCart(cart, &count, "Charger");
15
16     printf("Cart: %s", cart[0]);
17
18     return 0;
19 }
```

E:\06Aug_se\Practice\session- X + v

Cart: Charger

Process exited after 0.6268 seconds with return value 0

Press any key to continue . . .

Q 3 Write two functions: increase Followers By Value and increase Followers By Reference. Each should take a followers count and add 1000 to it, but one should use pass-by-value and the other pass-by-reference. Show how the original followers count changes (or doesn't) after calling each function.

Ans

```

Edit Search View Project Execute Tools AStyle Window Help
[Icons]
(globals)
on-8(1).cpp session-8(2).cpp session-8(3).cpp
#include <stdio.h>

void increaseByValue(int n)
{
    n = n + 1000;
}

void increaseByReference(int *n)
{
    *n = *n + 1000;
}

int main()
{
    int followers = 5000;

    increaseByValue(followers);
    printf("Value: %d\n", followers);

    increaseByReference(&followers);
    printf("Reference: %d", followers);

    return 0;
}

```

```

Value: 5000
Reference: 6000
-----
Process exited after 0.6302 seconds w
Press any key to continue . . .

```

Q 4. Build a function `format Price` that takes a price in rupees and returns a string formatted like Flipkart's price tag (e.g., '₹1,599'). Use this function to display prices for three different products.

Ans.

The image shows a screenshot of a Dev-C++ IDE window titled "E:\06Aug_se\Practice\session-8(4).cpp - [Executing] - Dev-C++ 5.11". The menu bar includes File, Edit, Search, View, Project, Execute, Tools, AStyle, Window, and Help. The toolbar contains various icons for file operations, execution, and search. The variable explorer on the right shows a global variable named "globals". The tab bar at the bottom displays four tabs: session-8(1).cpp, session-8(2).cpp, session-8(3).cpp, and session-8(4).cpp. The main editor area shows the source code of session-8(4).cpp, which includes a function `formatPrice` and a `main` function. The `main` function calls `formatPrice` with arguments 1599, 999, and 2499, and prints the results. The output window on the right shows the execution results: 1599, 999, and 2499, followed by a message "Process exited after Press any key to co".

```
#include <stdio.h>

char* formatPrice(int price)
{
    static char p[20];
    sprintf(p, "%d", price);
    return p;
}

int main()
{
    printf("%s\n", formatPrice(1599));
    printf("%s\n", formatPrice(999));
    printf("%s\n", formatPrice(2499));

    return 0;
}
```

1599
999
2499

Process exited after
Press any key to co

1599

999

2499

Q5. Refactor an existing function you wrote above to make it reusable for both product names and usernames (for example, a function that capitalizes the first letter of any string).

Constraint: The refactored function should work for any string input, not just a specific use case.

Ans.

E:\06Aug_se\Practice\session-8(5).cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

(globals)

session-8(1).cpp session-8(2).cpp session-8(3).cpp session-8(4).cpp session-8(5).cpp

```
1  #include <stdio.h>
2  #include <ctype.h>
3
4  void capitalize(char str[])
5  {
6      str[0] = toupper(str[0]);
7  }
8
9  int main()
10 {
11     char product[] = "mobile";
12     char username[] = "virat";
13
14     capitalize(product);
15     capitalize(username);
16
17     printf("Product: %s\n", product);
18     printf("Username: %s", username);
19
20     return 0;
21 }
```

E:\06Aug_se\Practice\session-

Product: Mobile
Username: Virat

Process exited after
Press any key to cont

Product: mobile

Username: virat

Session-9

Q 1. Declare a 1D array called daily Steps with 7 elements to store your step count for each day of the week, assign sample values, and print each value using a loop.

Ans.

E:\06Aug_se\Practice\session-9(1).cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

session-9(1).cpp

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int dailySteps[7] = {5000, 6000, 4500, 7000, 8000, 5500, 6500};
6
7      for(int i = 0; i < 7; i++)
8      {
9          printf("%d\n", dailySteps[i]);
10     }
11
12     return 0;
13 }
```

E:\06Aug_se\Practice\session-9(1).cpp

5000
6000
4500
7000
8000
5500
6500

Process exited after 0.3144 seconds with return value 0
Press any key to continue . . .

5000

6000

4500

7000

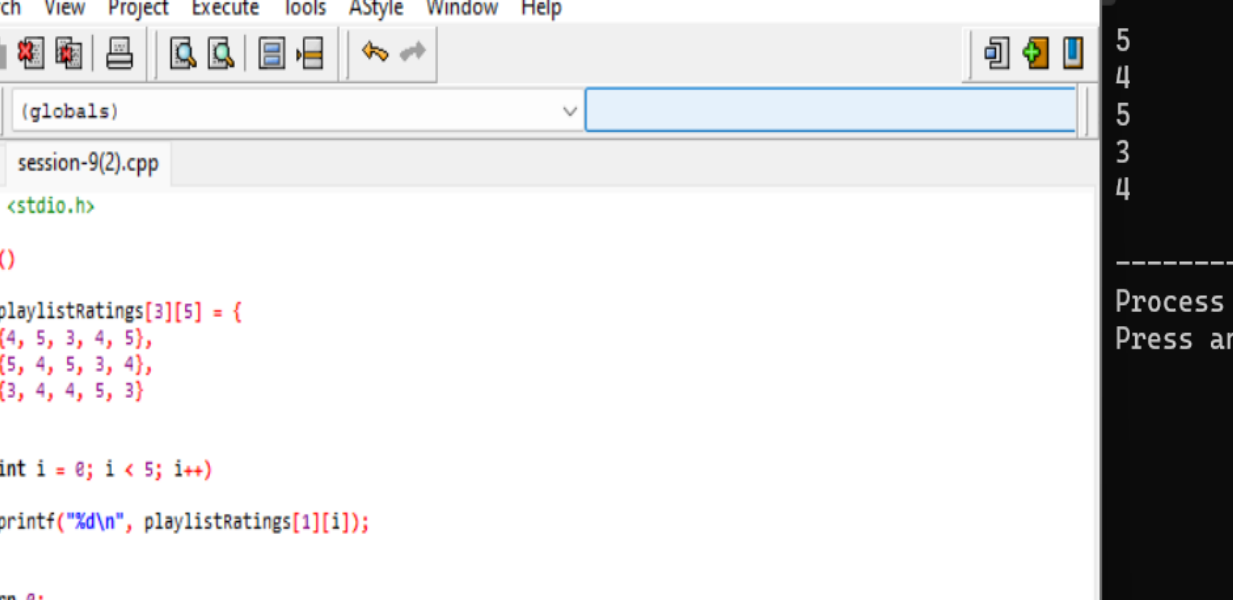
8000

5500

6500

Q.2 Create a 2D array called playlist Ratings to store ratings for 3 Spotify playlists over 5 days (rows = playlists, columns = days). Fill it with sample numbers and print the ratings for the second playlist.

Ans.



```
E:\06Aug_se\Practice\session-9(2).cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
[Icons]
(globals)
session-9(1).cpp session-9(2).cpp
1  #include <stdio.h>
2
3  int main()
4  {
5      int playlistRatings[3][5] = {
6          {4, 5, 3, 4, 5},
7          {5, 4, 5, 3, 4},
8          {3, 4, 4, 5, 3}
9      };
10
11     for(int i = 0; i < 5; i++)
12     {
13         printf("%d\n", playlistRatings[1][i]);
14     }
15
16     return 0;
17 }
```

5

4

5

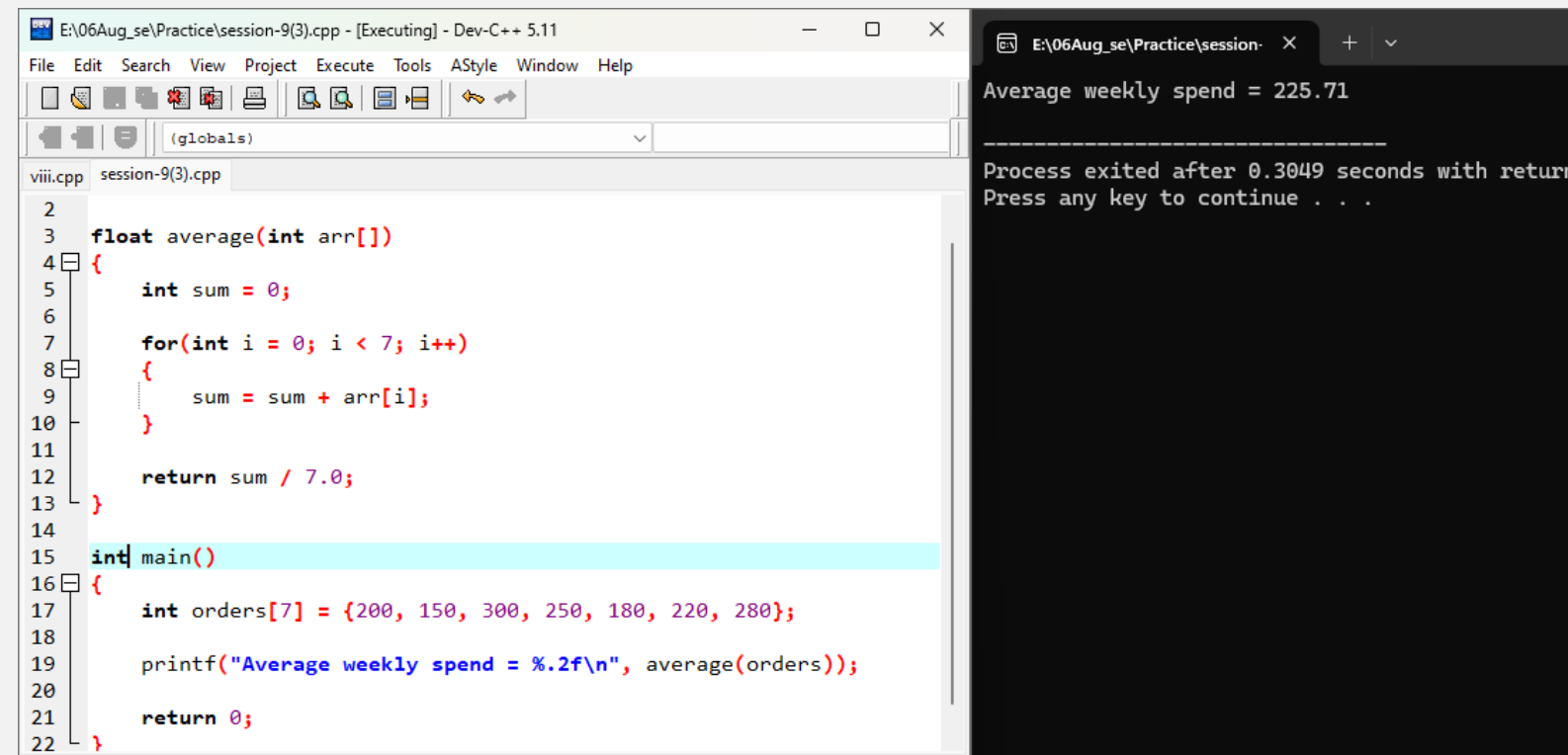
3

4

Q3. Build a function that takes a 1D array of 7 integers representing your daily Zomato order amounts and calculates the average spend for the week.

Hint: Use a loop to sum the values, then divide by the array length.

Ans.



The screenshot shows a C++ program in Dev-C++ 5.11. The program defines a function `average` that takes an integer array and returns its average. In the `main` function, an array `orders` with values {200, 150, 300, 250, 180, 220, 280} is passed to `average`, and the result is printed using `printf`.

```
2
3 float average(int arr[])
4 {
5     int sum = 0;
6
7     for(int i = 0; i < 7; i++)
8     {
9         sum = sum + arr[i];
10    }
11
12    return sum / 7.0;
13 }
14
15 int main()
16 {
17     int orders[7] = {200, 150, 300, 250, 180, 220, 280};
18
19     printf("Average weekly spend = %.2f\n", average(orders));
20
21     return 0;
22 }
```

The output window shows the execution result: "Average weekly spend = 225.71". Below this, it states "Process exited after 0.3049 seconds with return" and "Press any key to continue . . .".

Average weckly spend =225.71

Q 4. Given a 2D array called cricket Scores where each row represents an IPL match and columns represent runs scored by each team, write code to print the highest score from each match.

Ans.

E:\06Aug_se\Practice\session-9(4).cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

(globals)

viii.cpp session-9(3).cpp session-9(4).cpp

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int cricketScores[3][2] = {
6          {180, 160},
7          {150, 175},
8          {200, 190}
9      };
10
11     for(int i = 0; i < 3; i++)
12     {
13         if(cricketScores[i][0] > cricketScores[i][1])
14             printf("%d\n", cricketScores[i][0]);
15         else
16             printf("%d\n", cricketScores[i][1]);
17     }
18
19     return 0;
20 }
```

E:\06Aug_se\Practice\session-
180
175
200

Process exited after 0.6325 seconds with return
Press any key to continue . . .

180

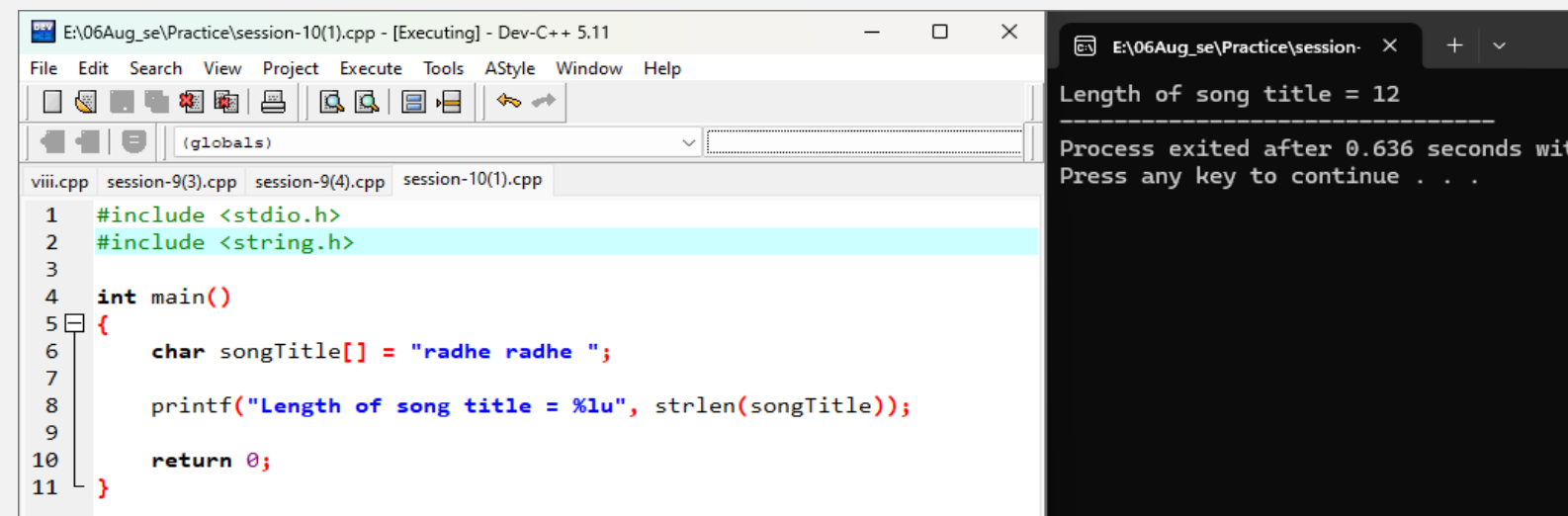
175

280

Session-10

Q 1. Declare a string variable called song Title and assign it the value 'Tum Hi Ho'. Print the length of the string using strlen ().

Ans.



The screenshot displays the Dev-C++ IDE interface. The main window shows a C++ program in `session-10(1).cpp`. The code includes `<stdio.h>` and `<string.h>`, defines a `main()` function, and declares a character array `songTitle` with the value "radhe radhe ". It uses `printf` to print the length of the string using `strlen` and then returns 0. The output window on the right shows the execution result: "Length of song title = 12", followed by a separator line and a message indicating the process exited after 0.636 seconds.

```
E:\06Aug_se\Practice\session-10(1).cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
viii.cpp session-9(3).cpp session-9(4).cpp session-10(1).cpp
1  #include <stdio.h>
2  #include <string.h>
3
4  int main()
5  {
6      char songTitle[] = "radhe radhe ";
7
8      printf("Length of song title = %lu", strlen(songTitle));
9
10     return 0;
11 }
```

Length of song title = 12

Process exited after 0.636 seconds with return code 0
Press any key to continue . . .

Length of song title=12

Q2. Ake input for two usernames (as strings) and compare them using `strcmp()`. Display whether they are the same or different.

Ans.

The screenshot displays the Dev-C++ IDE interface. The main window shows a C++ source file named `viii.cpp` with the following code:

```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char user1[20], user2[20];
7
8     printf("Enter username 1: ");
9     scanf("%s", user1);
10
11    printf("Enter username 2: ");
12    scanf("%s", user2);
13
14    if (strcmp(user1, user2) == 0)
15        printf("Same");
16    else
17        printf("Different");
18
19    return 0;
20 }
```

The output window on the right shows the execution results for two test cases:

Test Case 1:

```
Enter username 1: vishva
Enter username 2: vishva
Same
-----
Process exited after 7.251 seconds with return v
Press any key to continue . . .
```

Test Case 2:

```
Enter username 1: vishva
Enter username 2: mina
Different
-----
Process exited after 8.162 seconds with return v
Press any key to continue . . .
```

Enter username 1: Vishva

Enter username 1: mina

Different

Enter username 1: Vishva

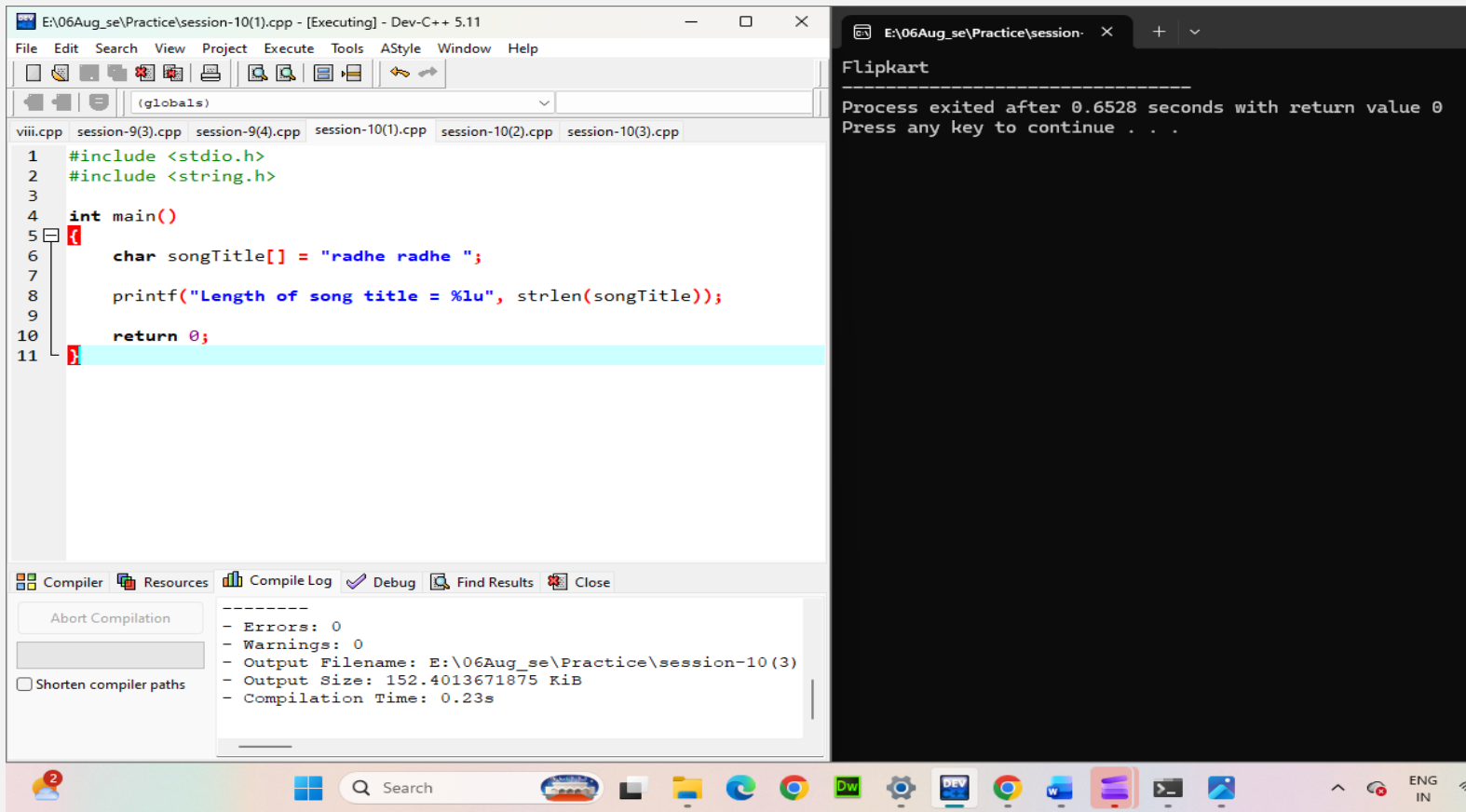
Enter username 1: Vishva

same

Q 3. Copy the string 'Flipkart' into another string variable called shopping App using strcpy (), then print the value of shopping App.

Hint: Make sure to declare enough space for the destination string.

Ans.

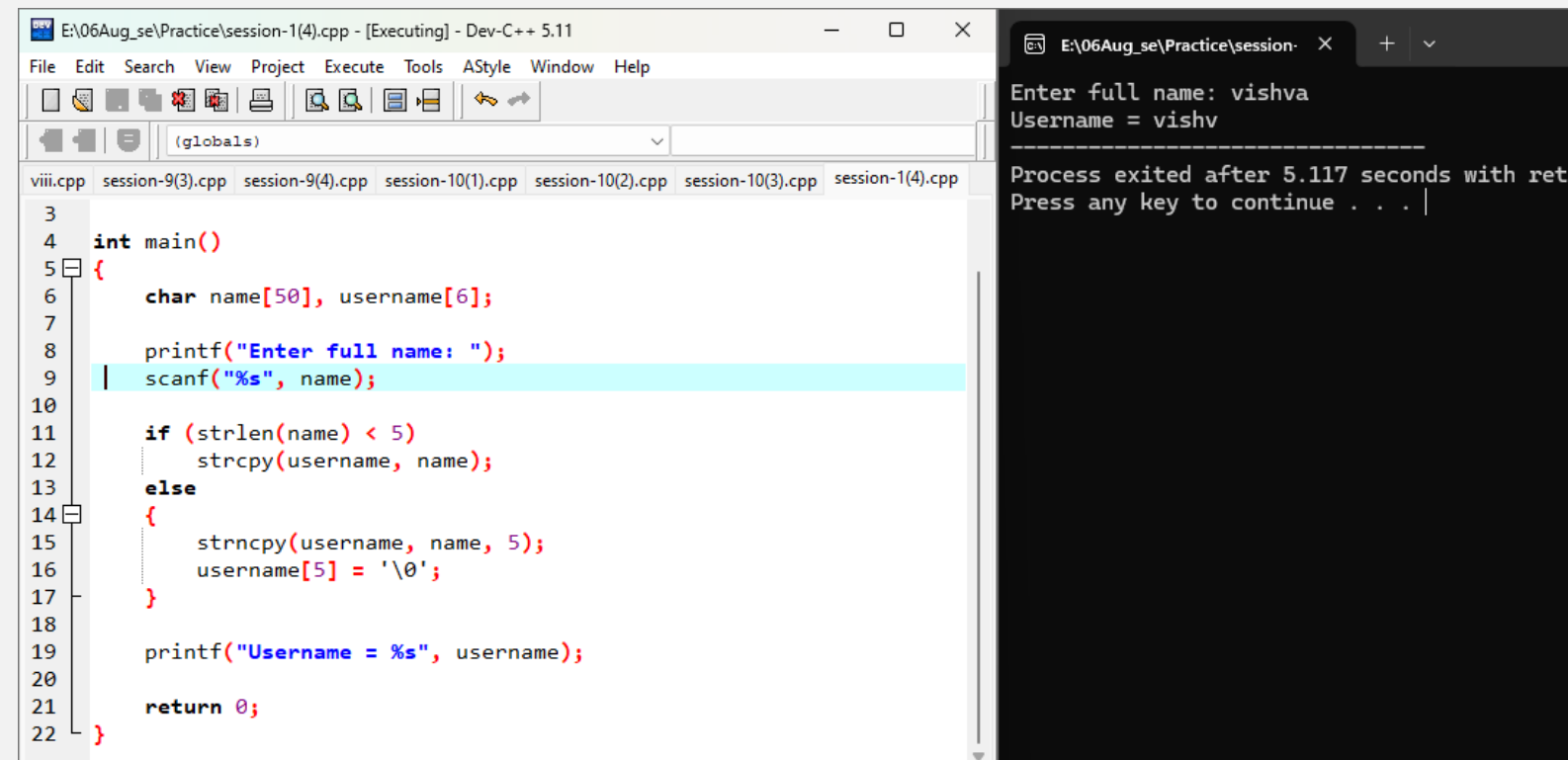


Filpkart

Q.4 Build a small script that takes the user's full name as input and creates a username by copying only the first 5 characters using strcpy (). Print the generated user name .

Constraint: If the name is shorter than 5 characters, use the full name as the user name.

Ans.



The screenshot shows a Dev-C++ IDE window titled "E:\06Aug_se\Practice\session-1(4).cpp - [Executing] - Dev-C++ 5.11". The code in the editor is as follows:

```
3
4  int main()
5  {
6      char name[50], username[6];
7
8      printf("Enter full name: ");
9      scanf("%s", name);
10
11     if (strlen(name) < 5)
12         strcpy(username, name);
13     else
14     {
15         strncpy(username, name, 5);
16         username[5] = '\0';
17     }
18
19     printf("Username = %s", username);
20
21     return 0;
22 }
```

The output window on the right shows the following text:

```
Enter full name: vishva
Username = vishv
-----
Process exited after 5.117 seconds with return code 0
Press any key to continue . . .
```

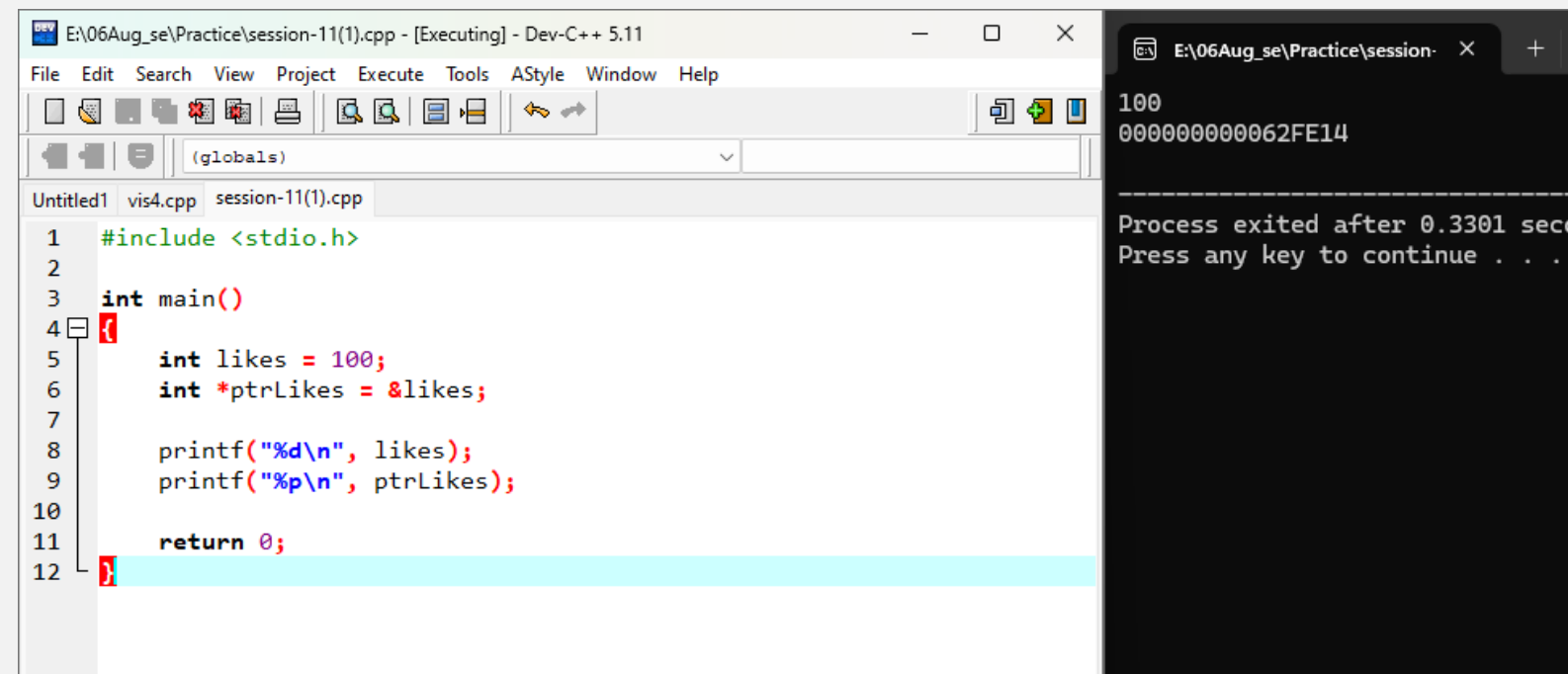
Enter full name = Vishva

Username=vishva

Session-11

Q 1. Declare an integer variable called likes and a pointer variable called ptrLikes; assign likes a value, point ptrLikes to likes, and print both the value and the address stored in ptrLikes.

Ans.



The screenshot displays the Dev-C++ IDE interface. The main window shows a C++ program named 'session-11(1).cpp' with the following code:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int likes = 100;
6     int *ptrLikes = &likes;
7
8     printf("%d\n", likes);
9     printf("%p\n", ptrLikes);
10
11     return 0;
12 }
```

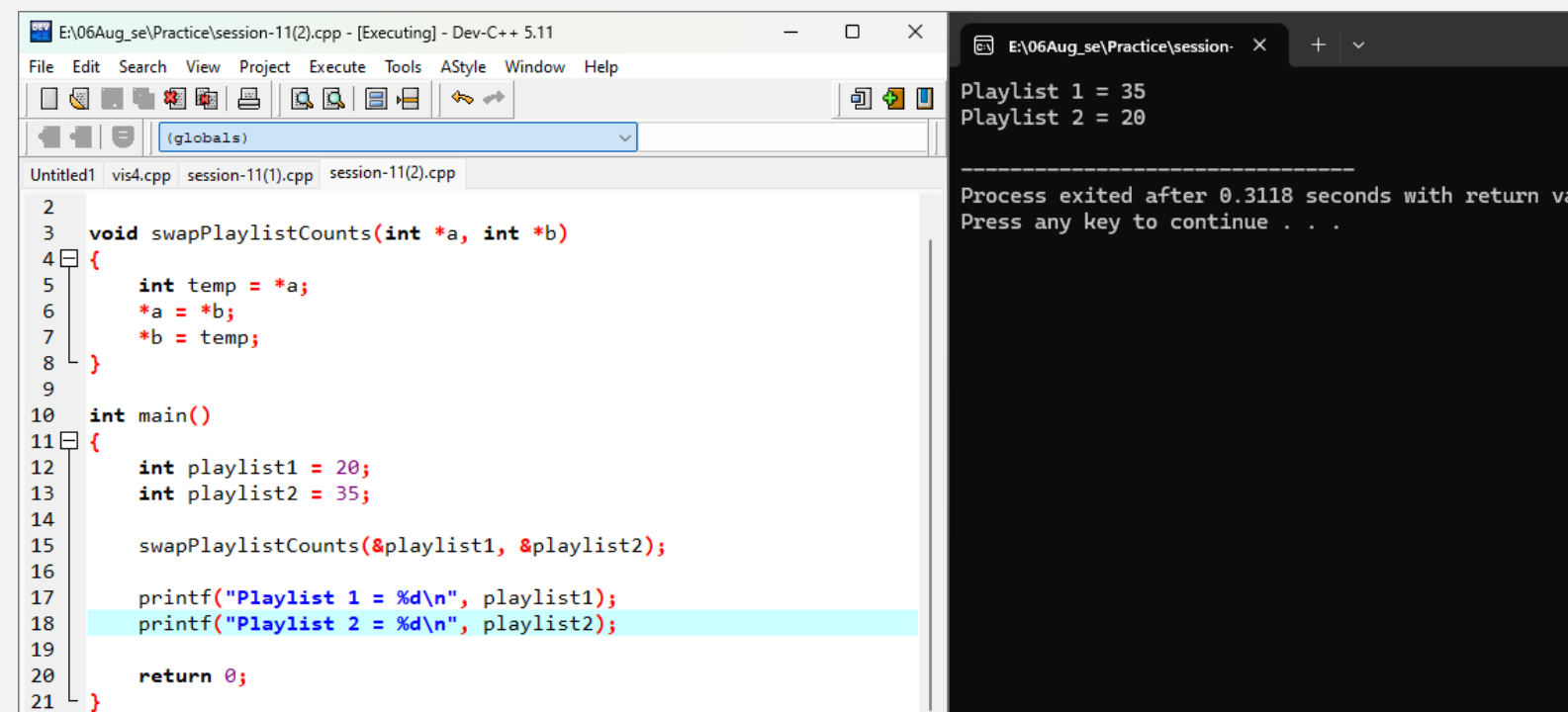
The output window on the right shows the execution results:

```
100
000000000062FE14

-----
Process exited after 0.3301 sec
Press any key to continue . . .
```

Q 2. Write a function swap Playlist Counts(int *a, int *b) that swaps the number of songs in two Spotify playlists using pointers, then call the function in main and print the swapped values.

Ans.



The screenshot displays a C++ program in the Dev-C++ IDE. The code defines a function `swapPlaylistCounts` that takes two integer pointers and swaps their values using a temporary variable. The `main` function initializes two variables, `playlist1` (20) and `playlist2` (35), calls the swap function with their addresses, and prints the values. The output window shows the result: `Playlist 1 = 35` and `Playlist 2 = 20`.

```
2
3 void swapPlaylistCounts(int *a, int *b)
4 {
5     int temp = *a;
6     *a = *b;
7     *b = temp;
8 }
9
10 int main()
11 {
12     int playlist1 = 20;
13     int playlist2 = 35;
14
15     swapPlaylistCounts(&playlist1, &playlist2);
16
17     printf("Playlist 1 = %d\n", playlist1);
18     printf("Playlist 2 = %d\n", playlist2);
19
20     return 0;
21 }
```

Playlist 1 = 35
Playlist 2 = 20

Process exited after 0.3118 seconds with return value 0
Press any key to continue . . .

Playlist 1 = 35

Playlist 2= 20

Q 3. Given an array of 5 order amounts (e.g., Zomato orders), use a pointer to iterate through the array and print each amount along with its memory address.

Hint: Use pointer arithmetic to move to the next element.

Ans.

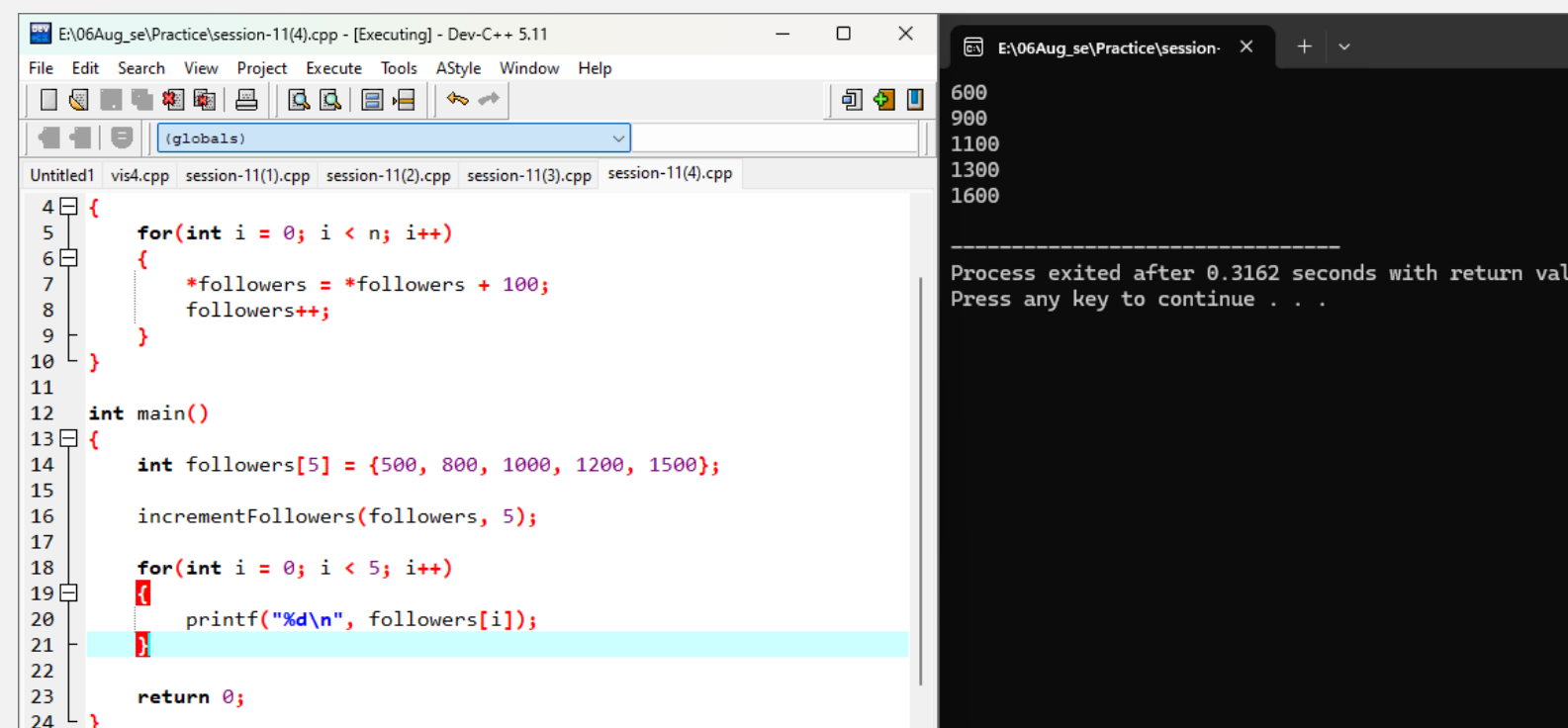
```
E:\06Aug_se\Practice\session-11(3).cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
(globals)
Untitled1 vis4.cpp session-11(1).cpp session-11(2).cpp session-11(3).cpp
1  #include <stdio.h>
2
3  int main()
4  {
5      int orders[5] = {100, 250, 150, 300, 200};
6      int *ptr = orders;
7
8      for(int i = 0; i < 5; i++)
9      {
10         printf("Amount = %d, Address = %radheshiyam\n", *ptr, (void *)
11             ptr);
12         ptr++;
13     }
14     return 0;
15 }
```

```
E:\06Aug_se\Practice\session-
Amount = 100, Address = radheshiyam
Amount = 250, Address = radheshiyam
Amount = 150, Address = radheshiyam
Amount = 300, Address = radheshiyam
Amount = 200, Address = radheshiyam

-----
Process exited after 0.3312 seconds with re
Press any key to continue . . .
```

Q 4. Create a function increment Followers (int *followers, int n) that increases each follower count in an array (representing Instagram followers for 5 friends) by 100 using pointer arithmetic, then print the updated counts.

Ans.



The screenshot shows a C++ IDE with a source code editor on the left and a console window on the right. The source code defines a function `incrementFollowers` that takes a pointer to an integer array and its size, and increments each element by 100. The `main` function initializes an array of 5 follower counts, calls the function, and prints the updated values.

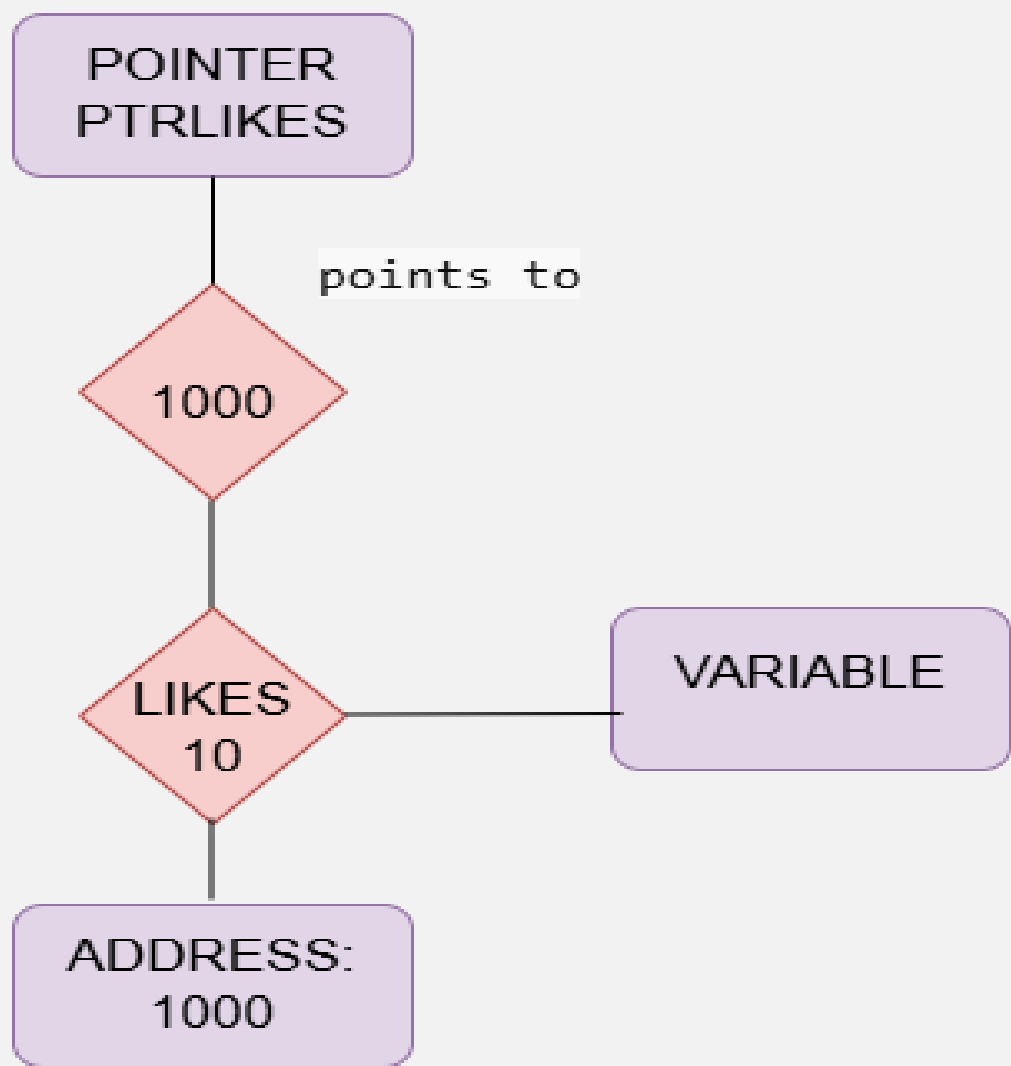
```
4 {  
5     for(int i = 0; i < n; i++)  
6     {  
7         *followers = *followers + 100;  
8         followers++;  
9     }  
10 }  
11  
12 int main()  
13 {  
14     int followers[5] = {500, 800, 1000, 1200, 1500};  
15  
16     incrementFollowers(followers, 5);  
17  
18     for(int i = 0; i < 5; i++)  
19     {  
20         printf("%d\n", followers[i]);  
21     }  
22  
23     return 0;  
24 }
```

The console window displays the output of the program, showing the updated follower counts for each of the 5 friends:

```
600  
900  
1100  
1300  
1600  
  
-----  
Process exited after 0.3162 seconds with return val  
Press any key to continue . . .
```

Q5. Draw a simple memory diagram (on paper or using any drawing tool) showing how a pointer references the address of a variable, using the variables from your likes and ptrLikes example; label the variable, pointer, and address.

Ans.



- likes → Variable
- ptrLikes → Pointer
- 1000 → Address